

A Taste of Single Variable Calculus

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Written using $\text{\LaTeX}$

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Preface

This text originally started off as a scattered compilation of review guides I made for my peers in AP Calculus BC to help them preview for unit tests. I soon realized it would be nice to put everything together into one cohesive document in hopes that it would help students first studying calculus or those reviewing calculus have an easier time. It would also serve as a nice personal project for me.

Since this text is introducing single variable calculus not necessarily from the most elementary axioms, students who have had absolutely no exposure to calculus or a shaky precalculus background may find it slightly difficult to navigate. But those who are comfortable with precalculus concepts or have had some exposure to calculus, or those who intend to use this as review material, should be comfortable navigating concepts in the text and understanding derivations.

There are numerous examples listed throughout the text that give some grounding to some concepts introduced throughout. While the solution is right underneath the question, you are still encouraged to try the question out yourself before consulting solutions. The best learning happens when you are frustrated and confused.

As briefly mentioned earlier, there are derivations scattered throughout the text pertaining to theorems you will come across in a first year calculus class. While some are necessary in understanding the intuition behind the theorems, some are simply “good to know”, and have been left in the appendix for those who are curious.

Calculus is an extremely versatile subset of mathematics that finds applications not only in further math, but all areas of science and even the more technical social sciences, like psychology or economics.

In Physics, it's used everywhere. Lagrangian mechanics uses calculus of variations, and statistical thermodynamics uses integrals to evaluate the Boltzmann distribution. In Chem-

istry, the rate of a reaction depends on reactant concentrations, forming an equation that has both a variable and its derivative. This is known as a differential equation. Spherical harmonics, which describe the behavior of molecular orbitals, adhere to Laplace's equation, which involves second derivatives. In Biology, predator prey systems are modeled by the Lotka-Volterra equations, a set of first order differential equations. Diffusion can also be modeled with derivatives. In psychology, integrals are used in statistical analysis to extract data from experiments. In Economics, slopes of curves can be modeled as derivatives of functions, and functions that depend on numerous variables can be analyzed with respect to each variable. There are more, and this goes to show how versatile calculus really is and how important it is to be equipped with this knowledge.

With this, we can begin our journey. We start with limits and continuity, concepts that one must know before diving into differential and integral calculus.

Chapter 1

Limits and continuity

1.1 Limit laws

Before we delve into further material, we must be sure of some limit laws. We will assume $\lim_{x \rightarrow a} f(x) = M$ and $\lim_{x \rightarrow a} g(x) = N$. Here they are:

$$\lim_{x \rightarrow a} [f(x) \pm g(x)] = M \pm N \quad (\text{Addition and subtraction.})$$

$$\lim_{x \rightarrow a} [k \cdot f(x)] = k M \quad (\text{Constant multiplication.})$$

$$\lim_{x \rightarrow a} [f(x) g(x)] = M N \quad (\text{Function multiplication.})$$

$$\lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{M}{N}, N \neq 0 \quad (\text{Division.})$$

$$\lim_{x \rightarrow a} [f(x)]^k = \left[\lim_{x \rightarrow a} f(x) \right]^k = M^k \quad (\text{Power.})$$

$$\lim_{x \rightarrow a} f(x)^{\frac{1}{k}} = M^{\frac{1}{k}} \quad (\text{Root.})$$

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right) = f(N) \quad (\text{Function in function.})$$

One relatively obvious condition to evaluate the limit is to check that it actually exists first. We rigorously say that

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

Otherwise, we cannot say that the limit is defined, and therefore cannot argue that it exists.

The squeeze theorem

This is a handy little theorem you can use to show a function must converge at some value given some conditions about adjacent functions. If $f(x) \leq g(x) \leq h(x)$ over some interval containing the point a , and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} g(x) = L$. One prerequisite to use this theorem is that all three functions are continuous. What does that mean?

1.2 Continuity

We say a function $f(x)$ is continuous at a point a if it satisfies three conditions:

1. $f(x)$ is defined,
2. $\lim_{x \rightarrow a} f(x)$ exists, and
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

Think of some examples of functions that fail at either step 1, 2, or 3 (consider $f(x) = \frac{x^4}{x^2}$. At $x = 0$, which conditions of continuity does $f(x)$ fail to satisfy?).

We can extend our definition of continuity at a point a to continuity over an interval $\mathcal{I}$. If $f(x)$ is continuous $\forall x \in \mathcal{I}$, then $f(x)$ is continuous over $\mathcal{I}$.

Types of discontinuities

There are three types of discontinuities we deal with.

Non-removable discontinuities exist when a function $f(x)$ tends toward $\pm\infty$ as $x \rightarrow a$. This implies $\lim_{x \rightarrow a} f(x) = \pm\infty$, which doesn't exist. An example is

$$\lim_{x \rightarrow 1} \frac{(x+2)^2}{(x-1)}.$$

Jump discontinuities exist when $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Good examples include piecewise functions.

Removable discontinuities exist when

$$\lim_{x \rightarrow a} f(x) = L, \quad \text{but} \quad f(a) \neq L.$$

A note on continuity of certain functions and composites

All polynomials are continuous everywhere, that is, they are continuous $\forall x \in \mathbb{R}$. This applies to rational functions (except, of course, at their points of discontinuity).

If $f(x), g(x)$ are both continuous functions at a , $f + g, f - g, cf, fg, \frac{f}{g}$ are all continuous, provided that $c \in \mathbb{R}$ and $g(a) \neq 0$. If f is continuous at b , and $\lim_{x \rightarrow a} g(x) = b$, then

$$\lim_{x \rightarrow a} (f \circ g)(x) = f(b).$$

The intermediate value theorem

It states that for continuous $f(x)$ over an interval $[a, b]$, and $N \in [f(a), f(b)]$, with $f(a) \neq f(b)$, $\exists c \in [a, b] \mid f(c) = N$. Be careful that this theorem only works if $f(x)$ is *continuous*.

1.3 The ε, δ definition of a limit

Mathematically, this definition says

$$\lim_{x \rightarrow c} f(x) = L \iff \forall \varepsilon > 0, \exists \delta > 0 \mid 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

If you can, find a direct relationship between ε and δ through substitution. This is usually easiest with polynomials. If you find that you get two values of δ when solving for ε , always choose the smaller δ , otherwise you risk having $|f(x) - L| > \varepsilon$.

Example: Suppose the power output of a cyclist is $\mathcal{W}_0$ watts and is governed by the equation

$$\mathcal{W} = C_1 v^2 + C_2, \quad \text{where } C_1 \text{ and } C_2 \text{ are positive constants.}$$

If we wish to keep the cyclist's power output to within ε watts of $\mathcal{W}_0$, by how much can the velocity vary?

Solution: We set up the bounded equation:

$$\begin{aligned}\mathcal{W}_0 - \varepsilon &\leq \mathcal{W} \leq \mathcal{W}_0 + \varepsilon \\ \mathcal{W}_0 - \varepsilon &\leq C_1 v^2 + C_2 \leq \mathcal{W}_0 + \varepsilon \\ \frac{\mathcal{W}_0 - \varepsilon - C_2}{C_1} &\leq v^2 \leq \frac{\mathcal{W}_0 + \varepsilon - C_2}{C_1} \\ \boxed{\sqrt{\frac{\mathcal{W}_0 - \varepsilon - C_2}{C_1}} \leq v \leq \sqrt{\frac{\mathcal{W}_0 + \varepsilon - C_2}{C_1}}}.\end{aligned}$$

Depending on values we choose for $\mathcal{W}_0, \varepsilon, C_2, C_1$, we can choose the expression that minimizes the difference between itself and v .

1.4 Deriving some special limits

We first aim to derive $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

First imagine a unit circle. We will draw a few areas:

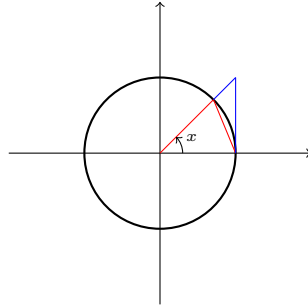


Figure 1.1: The unit circle set up.

Notice that the area covered by the red lines is smaller than the area of the circle spanned out by x , which in turn is smaller than the triangle in blue. Some trigonometry will tell you that the area of the red triangle is $\frac{\sin x}{2}$, the circle $\frac{x}{2}$, and the blue triangle $\frac{\tan x}{2}$. We now have a set of inequalities, which we'll manipulate.

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{\sin x}{2} &\leq \lim_{x \rightarrow 0} \frac{x}{2} \leq \lim_{x \rightarrow 0} \frac{\tan x}{2} \\
\lim_{x \rightarrow 0} \frac{\sin x}{\sin x} &\leq \lim_{x \rightarrow 0} \frac{x}{\sin x} \leq \lim_{x \rightarrow 0} \frac{\tan x}{\sin x} \\
\lim_{x \rightarrow 0} \frac{\sin x \cos x}{\sin x} &\leq \lim_{x \rightarrow 0} \frac{\sin x}{x} \leq \lim_{x \rightarrow 0} 1 \\
\lim_{x \rightarrow 0} \cos x &\leq \lim_{x \rightarrow 0} \frac{\sin x}{x} \leq \lim_{x \rightarrow 0} 1.
\end{aligned}$$

Conveniently, we can use the squeeze theorem here. We see that $\cos x \rightarrow 1$ as $x \rightarrow 0$, and 1 must approach 1 as $x \rightarrow 0$. Therefore, we can conclude that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

We would now like to evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$. We can take a less geometric approach here, relying on pure algebraic manipulation:

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} &= -\frac{(\cos x - 1)}{x} \\
&= -\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x + 1)}{x(\cos x + 1)} \\
&= -\lim_{x \rightarrow 0} \frac{-\sin^2 x}{x(\cos x + 1)} \\
&= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{\sin x}{\cos x + 1} \\
&= 1 \cdot \frac{0}{2} = 0.
\end{aligned}$$

Chapter 2

Differentiation

A **derivative** can be thought of as the instantaneous slope at any point in some function $f(x)$. Mathematically,

$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Some functions are differentiable everywhere. Some are not. What differentiates (pun intended) the two?

2.1 Is the function differentiable?

We describe a few conditions of differentiability. If $f(x)$ is differentiable at a , then $f'(a)$ must exist. Here, we use the Newton notation of a derivative. The derivative must not only exist at a , but $f'(a)$ must assume a unique value. That's why $\frac{d}{dx}|x|$ at $x = 0$ doesn't exist: there is no unique value for $\frac{d}{dx}|x|$ there.

As we did with continuity, we can extend the definition of differentiability from a point to an interval. If $f(x)$ is differentiable $\forall x \in \mathcal{I}$, then $f(x)$ is differentiable over $\mathcal{I}$. Be careful about $\mathcal{I}$, though. It must be open (why?).

Something particularly useful to remember is that if a function is differentiable, it is continuous. However, the contrary is not always true. Can you think of any functions that are continuous but not differentiable?

Non-differentiable functions

We use non-differentiable here when we are referring to functions that are not differentiable everywhere. This includes functions with cusps, noncontinuous functions, and piecewise

functions, among others. An example is

$$f(x) = \begin{cases} \sqrt{1-x^2} & \text{when } x \leq 1 \\ -\sqrt{1-(x-2)^2} & \text{when } x > 1 \end{cases} \quad \text{at the point } x = 1.$$

Fun fact: A function that is continuous everywhere but differentiable nowhere is the *Weierstrass function*, defined as

$$f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x), \text{ where } 0 < a < 1, b = 2n + 1, n \in \mathbb{Z}^+, \text{ and } ab > 1 + \frac{3}{2}\pi.$$

2.2 Rules of derivatives

Here we look at a few rules for derivatives.

Theorem (Power rule for derivatives).

$$\frac{d}{dx} x^n = nx^{n-1}.$$

Derivation.

We will use the limit definition of a derivative:

$$\frac{d}{dx} f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Using the binomial theorem,

$$\begin{aligned} \frac{d}{dx} x^n &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sum_{k=1}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} \\ &= \lim_{h \rightarrow 0} \sum_{k=1}^{n-1} \binom{n}{k} x^{n-k} h^{k-1} \\ &= \binom{n}{1} x^{n-1} \\ &= \boxed{nx^{n-1}} \quad \square. \end{aligned}$$

Theorem (Exponent rule).

$$\frac{d}{dx} C^x = \ln(C) C^x, C \in \mathbb{R}^+.$$

Derivation.

Again, we use the limit definition of a derivative:

$$\frac{d}{dx} C^x = \lim_{h \rightarrow 0} \frac{C^{x+h} - C^x}{h}.$$

We can factor out C^x since it won't be affected by the limit. It may also be useful to know that $\lim_{h \rightarrow 0} \frac{C^h - 1}{h} = \ln C$ (if you want, you can prove this using L'Hôpital's rule, which is covered later). Thus,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{C^{x+h} - C^x}{h} &= C^x \lim_{h \rightarrow 0} \frac{C^h - 1}{h} \\ &= \boxed{C^x \ln C} \quad \square. \end{aligned}$$

Theorem (Product and quotient rules).

$$\text{Product rule : } \frac{d}{dx} f(x) g(x) = f'(x)g(x) + f(x)g'(x)$$

$$\text{Quotient rule : } \frac{d}{dx} \frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

Derivation.

$$\begin{aligned} \frac{d}{dx} f(x) g(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) g(x+h) - f(x) g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) g(x+h) - f(x) g(x+h) + f(x) g(x+h) - f(x) g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)] g(x+h) + f(x) [g(x+h) - g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot \lim_{h \rightarrow 0} g(x+h) + f(x) \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \boxed{f'(x)g(x) + f(x)g'(x)} \quad \square. \end{aligned}$$

The quotient rule can be similarly derived.

Theorem (Chain rule).

$$\frac{d}{dx} (f \circ g)(x) = (f' \circ g)(x) g'(x).$$

Derivation.

$$\begin{aligned} \frac{d}{dx} (f \circ g)(x) &= \lim_{h \rightarrow 0} \frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(f \circ g)(x+h) - (f \circ g)(x)}{h} \cdot \frac{g(x+h) - g(x)}{g(x+h) - g(x)} \\ &= \lim_{h \rightarrow 0} \frac{(f \circ g)(x+h) - (f \circ g)(x)}{g(x+h) - g(x)} \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \end{aligned}$$

Letting $c = g(x+h) - g(x)$, the above becomes

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{(f \circ g)(x+c) - (f \circ g)(x)}{c} \cdot \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= \boxed{(f' \circ g)(x) g'(x)} \quad \square. \end{aligned}$$

2.3 Implicit differentiation

When equations pose too much difficulty to find a way to express y in terms of x and find a derivative, we must rely on implicit differentiation.

For instance, there is no simple way to isolate y from the equation $x^3 + y^3 = 6xy$ or to find its derivative. But using implicit differentiation,

$$\begin{aligned} \frac{d}{dx} x^3 + \frac{d}{dx} y^3 &= \frac{d}{dx} 6xy \\ 3x^2 + 3y^2 \frac{dy}{dx} &= 6(y + x \frac{dy}{dx}) \\ &= 6y + 6x \frac{dy}{dx} \\ 3x^2 - 6y &= (6x - 3y^2) \frac{dy}{dx} \\ \frac{dy}{dx} &= \boxed{\frac{3x^2 - 6y}{6x - 3y^2}} \end{aligned}$$

The chain rule is used to extract $\frac{dy}{dx}$.

2.4 Dealing with derivatives of other functions

Inverse trigonometric functions

What is $\frac{dy}{dx}$ of $y = \sin^{-1}(x)$? Think about it this way: if $y = \sin^{-1}(x)$, then $\sin(y) = x$. Now, we can perform implicit differentiation on both sides.

You should end up with something like $\cos(y)\frac{dy}{dx} = 1$. So, $\frac{dy}{dx} = \frac{1}{\cos(y)}$. But $\cos(y)$ is nothing more than $\sqrt{1 - \sin^2(y)}$. Remember that $\sin(y) = x$, so $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$.

Using similar methods, you can also find out $\frac{d}{dx} \cos^{-1}(x)$ and $\frac{d}{dx} \tan^{-1}(x)$ (try it!).

Inverse functions

If f is differentiable over interval $\mathcal{I}$, then $f^{-1}(x)$ is also differentiable over $\mathcal{I}$.

If $f^{-1}(x) = y$, then $f(y) = x$. Implicit differentiation yields $f'(y)\frac{dy}{dx} = 1$. Rearranging, we get that $\frac{dy}{dx} = \frac{1}{f'(y)} = \frac{1}{f'(f^{-1}(x))}$

Log functions

Here are two essential log derivatives you should know:

$$\frac{d}{dx} \log_b x = \frac{1}{x \ln b} \qquad \frac{d}{dx} \ln g(x) = \frac{g'(x)}{g(x)}.$$

Try deriving these formulas.

Logarithmic differentiation can really come in handy when you have very nasty equations to derive, such as

$$y = \frac{x^{\frac{3}{4}}(x^2 + 1)^{\frac{1}{2}}}{(3x + 2)^5}.$$

If you take the natural log of both sides, you get

$$\begin{aligned} \ln(y) &= \frac{3}{4} \ln(x) + \frac{1}{2} \ln(x^2 + 1) - 5 \ln(3x + 2) \\ \frac{1}{y} \frac{dy}{dx} &= \frac{3}{4} \left(\frac{1}{x} \right) + \frac{1}{2} \left(\frac{2x}{x^2 + 1} \right) - 5 \left(\frac{3}{3x + 2} \right) \\ \frac{dy}{dx} &= \frac{x^{\frac{3}{4}}(x^2 + 1)^{\frac{1}{2}}}{(3x + 2)^5} \left(\frac{3}{4} \left(\frac{1}{x} \right) + \frac{1}{2} \left(\frac{2x}{x^2 + 1} \right) - 5 \left(\frac{3}{3x + 2} \right) \right) \end{aligned}$$

Not a pretty result, but at least we've got something.

2.5 Related rates

There are some problems in calculus that involves derivatives that are related in some way, hence the name related rates. A lot of the time, some rates of change will be known, leaving you to solve for the unknown rates of change. They are usually related through geometry, so knowing some geometric properties will help. Because this isn't necessarily a concept, but rather a class of situations you might come across, we'll look at a few examples.

Example: A ladder with length 10 slides down a wall:

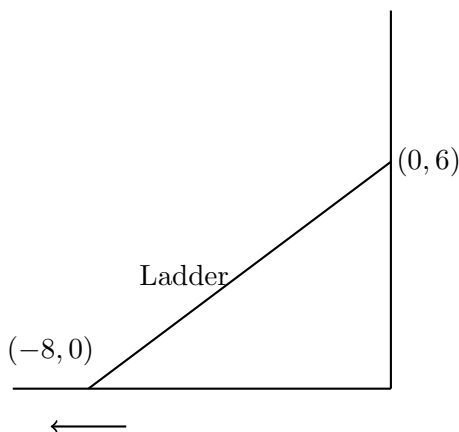


Figure 2.1: Our ladder at time t .

Its horizontal velocity is 2 meters per second to the left at some time t . What is its vertical velocity at that time?

We know that at time t , the horizontal velocity, $\frac{dx}{dt}$, is -2 meters per second. We also know that $x^2 + y^2 = 10^2$ must hold true at all times. We can now implicitly differentiate both sides to find that

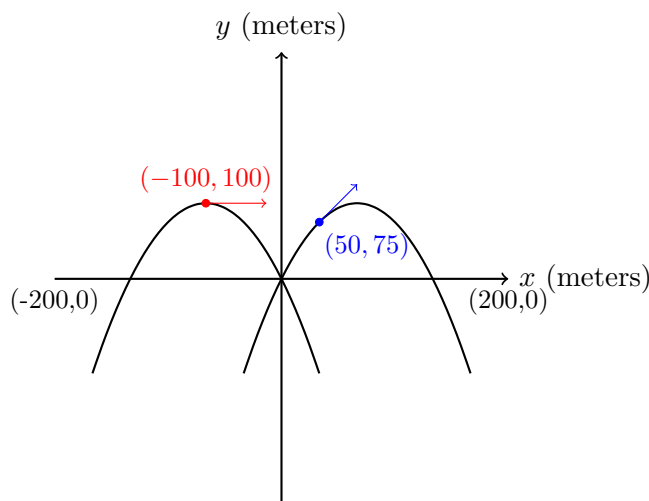
$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0, \quad \text{meaning} \quad x \frac{dx}{dt} = -y \frac{dy}{dt}.$$

We're also given the horizontal and vertical lengths of the ladder at time t , so we can use that to help us solve $\frac{dy}{dt}$. Doing algebra will give you that

$$\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt} = -\left(-\frac{8}{6}\right)(-2) = -\frac{8}{3} \text{ meters per second.}$$

Hence, the vertical velocity is decreasing at $\frac{8}{3}$ meters per second at time t .

Example: Suppose two cars are traveling in parabolic trajectories in a hypothetical country that has very oddly shaped roads. One car travels at 35 meters per second, while the other travels at 28 meters per second.



The car in blue is traveling at 28 meters per second, while the one in red 35 meters per second. How fast is their distance changing at this point? You can assume that both parabolic trajectories have a maximum at $(100, 100)$.

Here, we are given both $\frac{dx}{dt}$ and $\frac{dy}{dt}$. We would like to find $\frac{dD}{dt}$, where D is the distance between the two vehicles. We can use the distance formula:

$$D = \sqrt{(x_B - x_R)^2 + (y_B - y_R)^2},$$

where x_B is the blue car's x coordinate and x_R is the red car's x coordinate. Implicit differentiation gives us

$$\frac{dD}{dt} = \frac{1}{2D} \cdot \left(2(x_B - x_R) \left(\frac{dx_B}{dt} - \frac{dx_R}{dt} \right) + 2(y_B - y_R) \left(\frac{dy_B}{dt} - \frac{dy_R}{dt} \right) \right).$$

This means we just need to find the horizontal and vertical components of each vehicle; we already have their coordinates. $\frac{dx_R}{dt}$ and $\frac{dy_R}{dt}$ are decently easy, since the velocity vector is horizontal.

$$\frac{dx_R}{dt} = 35 \text{ meters per second}, \frac{dy_R}{dt} = 0 \text{ meters per second}.$$

The other vehicle is a bit more painful to figure out, but based on your knowledge of

parabolas, you can figure out that the slope, $\frac{dy}{dx}$, is 1. Hence, we can construct a right triangle to model the velocity vector. Using some special right triangle geometry, we find

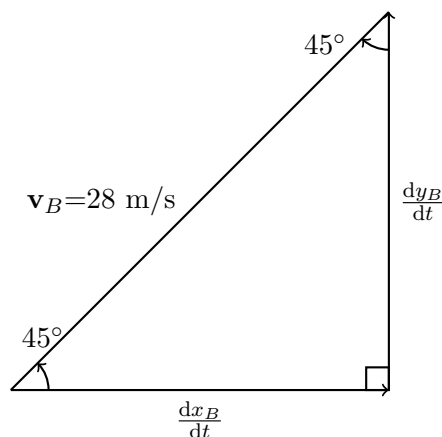


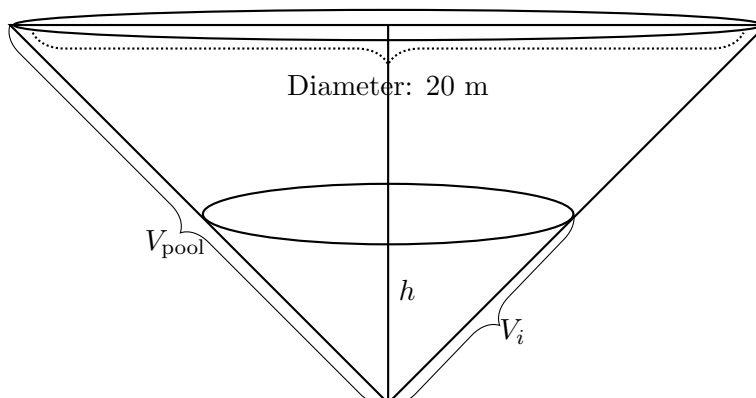
Figure 2.2: The velocity vector of the blue car, dissected.

that $\frac{dx_B}{dt}$ and $\frac{dy_B}{dt}$ are both $14\sqrt{2}$ meters per second. Now, we substitute all the values we have into our equation from earlier to find that

$$\frac{dD}{dt} = \frac{1}{152.069} \left((150) (14\sqrt{2} - 35) + (25) (14\sqrt{2}) \right) \text{ meters per second} = -11.745 \text{ m/s}.$$

Hence we can conclude that the cars are getting closer to each other at 11.745 meters per second at this moment.

Example: Suppose a conical swimming pool is being filled up with water. The pool's depth is 10 meters and the diameter at surface level is 20 meters. Water is being pumped into the pool at a rate of 20 liters per minute. When the pool is half filled, how fast is the water level rising?



The figure above hopefully visualizes what the pool looks like. Here, we would like to find $\frac{dh}{dt}$ when the pool is half filled. We also have information about the rate of water flow, $\frac{dV}{dt}$. This allows us to set up an equation relating the volume and some other quantities.

$$V = \frac{1}{3}\pi r^2 h \quad \implies \quad \frac{dV}{dt} = \frac{1}{3}\pi \left(2r \frac{dr}{dt} h + r^2 \frac{dh}{dt} \right).$$

While this equation initially looks daunting, it helps to realize that $h = r$ for any value we choose. This simplifies the equation significantly such that

$$\frac{dV}{dt} = \frac{1}{3}\pi \left(3h^2 \frac{dh}{dt} \right) = \pi h^2 \frac{dh}{dt}.$$

Since we know that $\frac{dV}{dt} = 20$, we now need to solve for h when $V_i = \frac{V_{\text{pool}}}{2}$, then solve for $\frac{dh}{dt}$.

$$\begin{aligned} V_i &= \frac{1}{3}\pi r_i^2 h_i \\ V_{\text{pool}} &= \frac{1}{3}\pi r_{\text{pool}}^2 h_{\text{pool}} \approx 1047.2 \text{ liters.} \\ \implies \frac{1}{3}\pi r_i^2 h_i &= 523.6 \text{ liters.} \\ \text{but since } r &= h, \frac{1}{3}\pi h_i^3 = 523.6 \text{ liters.} \\ \implies h_i &= \sqrt[3]{\frac{523.6 \cdot 3}{\pi}} = 7.94 \text{ meters.} \end{aligned}$$

Now we have everything and can solve for $\frac{dh}{dt}$.

$$\begin{aligned} \frac{dV}{dt} &= \pi h^2 \frac{dh}{dt} \\ 20 &= \pi (7.94)^2 \frac{dh}{dt} \\ \implies \frac{dh}{dt} &= \frac{20}{\pi (7.94)^2} \approx 0.1 \text{ meters per minute.} \end{aligned}$$

We see that when the volume of the water in the pool is half that of the pool, the water level rises at 0.1 meters per minute. These three examples are situations where related rates are important, but can be a lot more variation in situations. You'll also notice how all of our answers have units on them. Pay close attention to the units you're working with and make sure you always double check to see if your units make sense. For instance, if you arrive at a change of distance between two objects in units of meters squared per liter, you know you've made a mistake somewhere.

2.6 Linear approximation

Linear approximations are important tools in mathematics and sciences. For equations that are otherwise difficult or impossible to analyze, creating the appropriate linear approximation at a certain point can vastly simplify the equation or provide solutions that are approximately equal to an otherwise unsolvable analytic solution. An equation that shows up in mechanics is the pendulum equation:

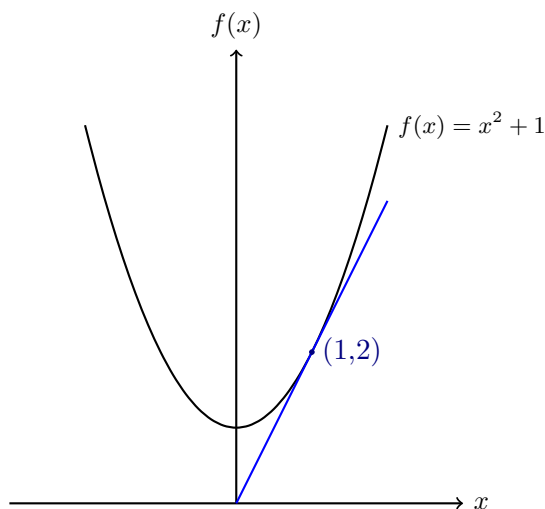
$$\frac{d^2x}{dt^2} + \frac{g}{\ell} \sin x = 0.$$

This is called a *nonlinear second order differential equation*, and analytic solutions are extremely hard to derive. However, a linear approximation for $\sin x$ near $x = 0$ is $f(x) = x$, making the equation much simpler:

$$\frac{d^2x}{dt^2} + \frac{g}{\ell} x = 0.$$

The equation is now known as a linear second order differential equation and has solutions far easier to find. This is known as the *small angle approximation*. Let us now see how we can arrive at the formula for such an approximation.

Let's begin by choosing any arbitrary function, $f(x)$. In the figure below, I've chosen $f(x) = x^2 + 1$, but the principles that I'll use to derive the linear approximation can apply to any differentiable function $f(x)$. We'll denote the linear approximation as $L(x)$.



In our figure, we see the blue line is $L(x)$, going through our graph at $(1, 2)$. This is the linear approximation of $x^2 + 1$ near the point $x = 1$. Let's look at some properties of $L(x)$. First, it appears as though the linear approximation is tangent to our function at the

point we'd like to approximate, meaning the slope of $L(x) = f'(a)$ for an approximation around $x = a$. Secondly, it goes through the point $(a, f(a))$. Using this information, we can already construct a linear approximation. Recall the point-slope form of a line:

$$y - y_i = m(x - x_i).$$

The point (x_i, y_i) that $L(x)$ goes through is simply $(a, f(a))$, and its slope must be $f'(a)$. Substituting these values back into our point-slope form of a line, we get

$$y - f(a) = f'(a)(x - a), \text{ or } L(x) = f(a) + f'(a)(x - a).$$

This is exactly the form we want. For our figure above,

$$L(x) = 2x.$$

For our sine function, we see that

$$L(x) = \sin(0) + \cos(0)x = x.$$

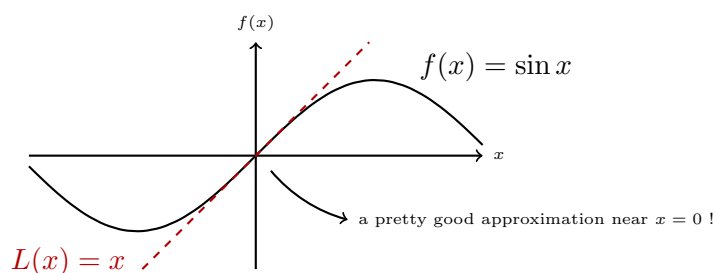


Figure 2.3: The linear approximation for $f(x) = \sin x$.

Chapter 3

Applications of derivatives

In this chapter, we'll see why derivatives are so important. They have far-reaching applications in most things you can graph (think stock prices, velocity-time graphs, etc.).

3.1 Extrema

Consider a point $c \in D$, where D is the domain of $f(x)$. We say that c is an *absolute maximum* if $\forall x \in D, f(c) \geq f(x)$. Similarly, we say c is an *absolute minimum* if $\forall x \in D, f(c) \leq f(x)$.

Now consider an interval I , which must lie in D . For a point $c \in I$, if $f(c) \geq f(x) \forall x \in I$, c is a *local maximum*. It follows that if $f(c) \leq f(x) \forall x \in I$, c is a *local minimum*.

We now look at a few theorems regarding extrema of functions.

The **Extreme Value Theorem** tells us that if f is continuous along the interval $[a, b]$, f must have a local minimum and maximum at some points c, d within $[a, b]$. Why does this not apply if we consider the interval (a, b) ?

Another proposition is **Fermat's theorem**: If $f(c)$ is a local extrema, and $f'(c)$ exists, then $f'(c) = 0$. Is the converse of Fermat's theorem true? On this topic, we consider *critical numbers*, elements c of the domain such that $f'(c) = 0$ or doesn't exist.

This is all great, but how would we find these local extrema? Over our open interval $[a, b]$, first find all critical values $c \in [a, b]$, then evaluate $f(a)$ and $f(b)$. Gather these values, and find your minima and maxima.

One last theorem (for this section, at least)! If $f(x)$ is *continuous* over $[a, b]$, *differentiable* over (a, b) , and $f(a) = f(b)$, $\exists c \in [a, b] \mid f'(c) = 0$. This is known as **Rolle's theorem**.

3.2 First derivatives and functions

One of the more important propositions in calculus is the **Mean Value Theorem**. Let $f(x)$ be continuous over $[a, b]$ and differentiable over (a, b) . Then

$$\exists c \in [a, b] \mid f'(c) = \frac{f(b) - f(a)}{b - a}, \quad \text{or} \quad f(b) - f(a) = f'(c)(b - a).$$

There are numerous applications for this, as we will see.

Meanwhile, we will discuss another application of derivatives. You've probably been asked to draw graphs of functions before, and while some are easy to draw, some may be much harder or impossible to visualize. Here's where calculus helps.

Suppose we have a funky function like $f(x) = x(x - 6)^{\frac{1}{3}}$. If we wanted to sketch this with no additional knowledge, it would be quite difficult. However, let's take the derivative of the function and set up a *sign chart*. This chart will tell us about critical values and give us a general idea of the graph. We find that

$$f'(x) = \frac{2(2x - 9)}{3(x - 6)^{\frac{2}{3}}}, \text{ meaning our critical values } c \text{ are } x = \frac{9}{2}, 6.$$

Our sign chart looks something like this:

Interval setup	f'
$(-\infty, \frac{9}{2})$	—
$\frac{9}{2}$	0
$(\frac{9}{2}, 6)$	+
6	DNE
$(6, \infty)$	+

Given this, try sketching $f(x)$! While sketching, you may notice some things. In general, if $\text{sgn}f'(x_i) \neq \text{sgn}f'(x_j)$, $x_i < c$, $x_j > c$ for a critical number c , you have a local extrema. In particular, if $\text{sgn}f'(x_i) = 1$ but $\text{sgn}f'(x_j) = -1$, you have a local maximum, and if $\text{sgn}f'(x_i) = -1$ but $\text{sgn}f'(x_j) = 1$, you have a local minimum.

How do we tell whether a function is increasing or decreasing ($f'(x) > 0$, $f'(x) < 0$)? We can test it. Suppose $f(x)$ is continuous over an interval I and we have two points $x_1, x_2 \in I$. If $x_1 \rightarrow x_2 \implies f(x_1) < f(x_2)$, f is increasing, but if $x_1 \rightarrow x_2 \implies f(x_1) > f(x_2)$, then f is decreasing. Be careful of the limitations: We don't know the exact behavior of $f(x)$ between x_1 and x_2 , we only know its general behavior between the two points. If you wanted to further analyze the function, you could set up a sign chart for that interval.

3.3 Second derivatives and their implications

So far, we've been looking extensively at first derivatives. But second derivatives are also very important. They can tell us whether a function is concave up or down. Recall that a concave up function is one where any tangent line you draw to the function always underestimates neighboring points' values, and a concave down function where any tangent line overestimates neighboring values.

In the language of calculus, our function $f(x)$, when concave up, always has $f'(x)$ that is increasing; that means $f''(x) > 0$. However, if $f(x)$ is concave down, $f'(x)$ is always decreasing, meaning $f''(x) < 0$. The point where $f''(x)$ changes sign has a special name: the *inflection point*.

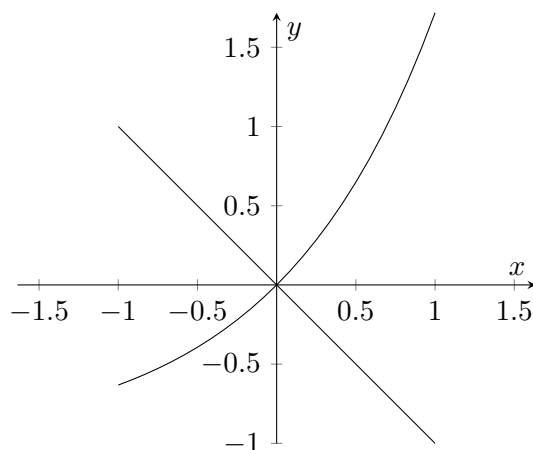
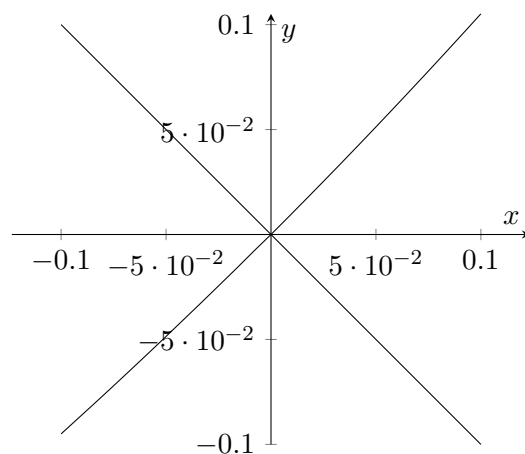
3.4 Indeterminate forms and L'Hôpital's rule

Some limits just can't be evaluated with tools we had before learning derivatives. In these cases, we'd have to use L'Hôpital's rule. HOWEVER, you must satisfy one of two indeterminate forms in order to use it: $\frac{\infty}{\infty}$, and $\frac{0}{0}$.

L'Hôpital's rule states that

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

Let's say we want to evaluate $\lim_{x \rightarrow 0} \frac{(e^x - 1)}{x}$. By noncalculus methods, this can't be easily evaluated. We will now see why L'Hôpital's rule works in this case:

Figure 3.1: The graphs of $e^x - 1$ and $-x$ over the domain $[-1, 1]$.Figure 3.2: The graphs of $e^x - 1$ and $-x$ over the domain $[-0.1, 0.1]$.

In the second figure, we see that the two graphs have essentially become lines, implying their derivatives around our point $x = 0$ are essentially constant, and the ratio of their distances from the x -axis remain constant in the neighborhood of $x = 0$.

Graphically, it appears that $e^x - 1$ has a slope of 1, while $-x$ has a slope of -1 around $x = 0$ (taking the derivative will confirm this). Both graphs race toward 0 at constant rates. Since the steeper graph will have a greater change over the same interval, it must start at a distance proportional to its rate of change to reach 0 at the same time as the less steep graph. This highlights how functions with larger derivatives near the point we wish to evaluate will be proportionally further away from the x -axis, and helps us understand the equality $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$.

Chapter 4

Antiderivatives – an introduction to integration

4.1 Antiderivatives and integrals

So far, we have explored one side of calculus: derivatives. We have looked at what it means for a function to have a derivative, how to take them, and some heuristics and rules. Now, we want to see what happens when we go backwards. As the name suggests, antiderivatives “reverse” what derivatives do.

A set of functions $\mathcal{A}$ is called the *antiderivative* of some function $f(x)$ if

$$\mathcal{A} = \left\{ A(x) \mid \frac{d}{dx} A(x) = f(x) \right\}.$$

You might be wondering why we say $\mathcal{A}$ is a family of functions. Consider the function $f(x) = x$. Which function’s derivative yields x ? One solution is $\frac{x^2}{2}$.

But $\frac{x^2}{2} + 1$ also works. So does $\frac{x^2}{2} + 2$, or $\frac{x^2}{2} + C, C \in \mathbb{R}$. Since all these functions have x as their derivative, they must fall into the same family of functions, $\mathcal{A}$.

There is a symbol in math we use to express the antiderivative of a function. The antiderivative can be expressed like an elongated S. This is also the same symbol we use for an **integral**.

Definite and indefinite integrals

We briefly explored antiderivatives and integrals above. There are a few differences and similarities to note between these two entities.

First and foremost, an antiderivative is synonymous with an indefinite integral. So, when we say we want the indefinite integral of x , that is the same thing as saying we want the antiderivative of x . Mathematically speaking, the indefinite integral of x looks like

$$\int x \, dx = \frac{x^2}{2} + C, C \in \mathbb{R}.$$

Notice how there is a dx attached to the end of the antiderivative. We will explain the notation behind this a little later.

We now look at definite integrals. They are somewhat similar to indefinite integrals, but instead of returning a family of functions, you end up with a number. We will differentiate the two as we explore more of integration.

4.2 Riemann sums and definite integration

You may have heard of an integral being described as the “area underneath the curve” for some function $f(x)$. This is partly true. What this quotation is describing, however, is a way to think about the definite integral, and can’t be generalized to all integrals. There are also a number of other nuances that the phrase doesn’t capture adequately, as we’ll find out soon.

How do we evaluate a definite integral, though? We can follow in the spirit of finding the area underneath the curve, but we have to be careful. Taking any area underneath the curve is problematic because we don’t know where to stop; the area underneath any curve is technically infinite.

In reality, we measure the area between the curve and the x axis. If the curve has $f(x) > 0$, we say it contributes positively to the integral, whereas if the curve has $f(x) < 0$, we say it contributes negatively.

Defining Riemann sums and the definite integral

Now that we have a better idea of what area to find, we want a way to actually do it. Curves like x are friendly in that they’re just a diagonal line, and finding the area requires

a bit of elementary geometry. However, most curves are not nearly as easy to evaluate.

The philosophy behind Riemann sums is to approximate the area of the curve $f(x)$ as a series of rectangles. Rectangles are simple geometric shapes that we can easily sum. We want to find an expression for these rectangles to sum.

Let's check out a visual example. Take an arbitrary function $f(x)$ and label two points you are interested in finding the area over. We'll call them a and b , with $a < b$. Now chop the function into n partitions between a and b . At each partition's leftmost point, extend a line perpendicular to the x axis until it touches $f(x)$. For the k th partition, we should have something like this:

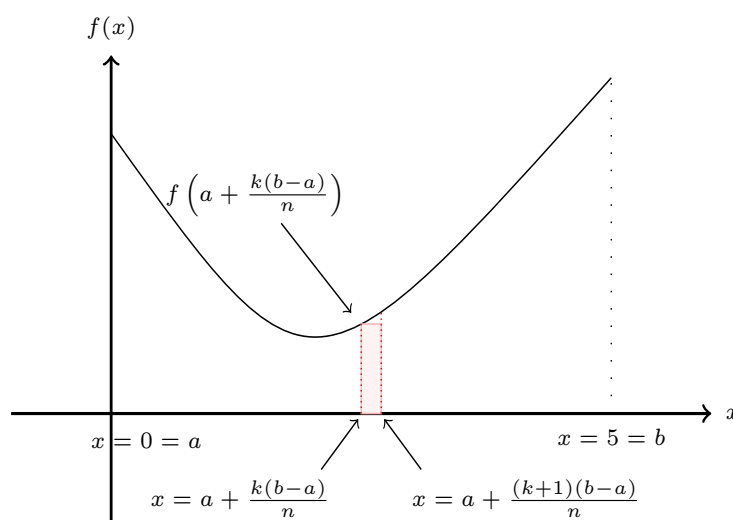


Figure 4.1: One particular rectangle in our partitions for some function $f(x)$.

We make such rectangles for all of our partitions, ranging from a to b . We also notice that each partition's horizontal length is $\frac{b-a}{n}$, which we'll rewrite as Δx . Piecing everything together, we find that the sum of our rectangles will approximate the area of $f(x)$, which can be written mathematically:

$$\text{Area}_{k \rightarrow k+1} = f\left(a + \frac{k(b-a)}{n}\right) \Delta x, \quad \text{Area } f(x)_{a \rightarrow b} \approx \sum_{k=0}^n \text{Area}_{k \rightarrow k+1} = \sum_{k=0}^n f\left(a + \frac{k(b-a)}{n}\right) \Delta x.$$

What we have here is known as a **Left-handed Riemann sum**.

There is also a **Right-handed Riemann sum**, defined quite similarly:

$$\text{Area } f(x)_{a \rightarrow b} \approx \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \Delta x.$$

Notice, however, that we start with $k = 1$ instead of $k = 0$.

We can now look at the **midpoint sum**, which takes the averages of a left and right handed Riemann sum.

Finally, we will explore the **trapezoidal sum**, which does pretty much what it sounds like it does. Instead of rectangles, we use trapezoids whose bases are the left and right sides of each partition, resulting in a trapezoid formula as follows:

$$A_t = \frac{1}{2} (b_1 + b_2) h = \frac{1}{2} \left(f(x) + f\left(x + \frac{b-a}{n}\right) \right) \Delta x.$$

Summing over all partitions from a to b , we get

$$A = \frac{\Delta x}{2} \left(f(a) + 2 \left[\sum_{k=1}^{n-1} f\left(a + \frac{k(b-a)}{n}\right) \right] + f(b) \right).$$

These are some nice sums. What happens as we take n to increase without bound? It of course depends on the function, but the general notation we use is to swap out Δx with dx and the summation symbol with our elongated “S”. This changes our finite sum into a definite integral!

$$\lim_{n \rightarrow \infty} \text{Area}_{k \rightarrow k+1} = \lim_{n \rightarrow \infty} \sum_{k=0}^n f\left(a + \frac{k(b-a)}{n}\right) \Delta x = \int_a^b f(x) dx.$$

4.3 The fundamental theorem of calculus

This theorem underpins all of calculus and relates integration to differentiation. There are a few direct consequences of it, which we'll briefly have a look at.

The theorem is stated as follows:

$$\text{If } \frac{d}{dx}F(x) = f(x), \text{ then } \int_a^b f(x) dx = F(b) - F(a).$$

A little something extra for the mathematically curious: We can also express the above as

$$dF = f(x) dx \implies \int_a^b f(x) dx = \int_I f(x) dx = \int_I dF = \int_{\partial I} F = F(b) - F(a).$$

The symbol ∂I is just the boundary/endpoints of our interval $I, \{a, b\}$. This result is generalizable to any number of dimensions, and is known as the *Generalized Stokes' theorem*.

Anyway, one of the consequences that arises is somewhat related. A function $g(x)$ can be represented as an area function from a to x of a separate function $f(t)$. This assumes $f(t)$ is continuous along an interval $I = [a, b]$ and $x \in I$. Symbolically, $g(x) = \int_a^x f(t) dt$. We also have a visual representation, shown below.

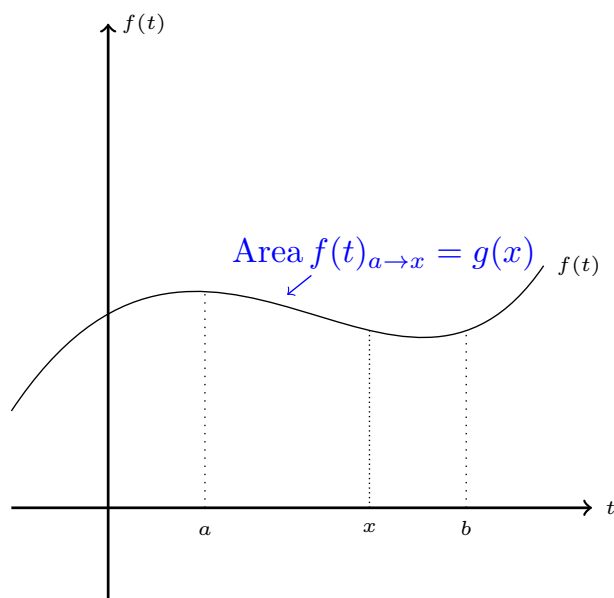


Figure 4.2: A visual representation of $f(t)$ parametrizing $g(x)$.

Taking derivatives of integrals

Notice something interesting about $g(x)$ that we described above. If we let $\frac{d}{dx}F(x) = f(x)$,

$$\begin{aligned}\frac{d}{dx} \int_a^x f(t) dt &= \frac{d}{dx} (F(x) - F(a)) \\ &= \frac{d}{dx} F(x) - \frac{d}{dx} F(a) \\ &= f(x).\end{aligned}$$

What happens if we're faced with something a little harder? We can let $f(t)$ parametrize a function $g(x)$ with more complexity. In full generality,

$$g(x) = \int_{p(x)}^{q(x)} f(t) dt.$$

How might we go about finding the derivative of such a function? Using the chain rule, this isn't so daunting after all:

$$\begin{aligned}\frac{d}{dx} g(x) &= \frac{d}{dx} \left(\int_{p(x)}^{q(x)} f(t) dt \right) \\ &= \frac{d}{dx} F(q(x)) - \frac{d}{dx} F(p(x)) \\ &= \frac{d}{dq(x)} F(q(x)) \frac{dq(x)}{dx} - \frac{d}{dp(x)} F(p(x)) \frac{dp(x)}{dx} \\ &= \boxed{f(q(x)) q'(x) - f(p(x)) p'(x)}.\end{aligned}$$

Average values

Integrals can also be quite useful in finding the average value of a function over an interval $I = [a, b]$. For n subintervals, the average value is expressed as

$$\overline{f(x)} = \frac{1}{n} \sum_{i=0}^n f \left(a + \frac{i(b-a)}{n} \right).$$

As we take n to increase without bound, we can find the average value for a continuous, integrable function. We'll relate it to a definite integral:

$$\begin{aligned}
\int_a^b f(x) \, dx &= \lim_{n \rightarrow \infty} \sum_{i=0}^n f\left(a + \frac{i(b-a)}{n}\right) \left(\frac{b-a}{n}\right) \\
&= (b-a) \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{1}{n} f\left(a + \frac{i(b-a)}{n}\right) \\
&= (b-a) \overline{f(x)} \\
\therefore \overline{f(x)} &= \frac{1}{b-a} \int_a^b f(x) \, dx.
\end{aligned}$$

4.4 Some handy integration tips

Just like differentiation, there are some shortcuts you can use to integrate without having to perform Riemann sums every time.

Theorem (Power rule for integration).

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad \text{and} \quad \int A^x \, dx = \frac{A^x}{\ln A} + C.$$

Other theorems will be introduced as we take a deeper dive into integration, but these are good to know for now.

We also want to look at a handy substitution trick to use when we have functions that are incredibly difficult to integrate otherwise. This is known as the method of ***u* substitution**.

The basic philosophy is as follows: Let $u = g(x)$ be a differentiable function over $I = [a, b]$. If $f(x)$ is continuous over I , then

$$\int_a^b f(g(x)) g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

See if you can make sense of why this is. This substitution makes the function we want to integrate, the **integrand**, simpler. We check a few examples.

Example 1: What is $\int_2^5 e^{-x^2} x \, dx$?

We perform a u substitution, letting $u = x^2$. Taking the derivative, we get that $du = 2x \, dx$,

which allows us to rewrite the integral

$$\int_2^5 e^{-x^2} x \, dx = \frac{1}{2} \int_4^{25} e^{-u} \, du = \boxed{\frac{1}{2} [e^{-4} - e^{-25}]}.$$

Example 2: Evaluate $\int_0^2 x^2 \cos(x^3) \, dx$.

This time, we use $u = x^3$, $\implies du = 3x^2 \, dx$. The integral becomes

$$\int_0^2 x^2 \cos(x^3) \, dx = \frac{1}{3} \int_0^8 \cos(u) \, du = \boxed{\frac{1}{3} \sin(8)}.$$

Example 3: Evaluate $\int x^5 \sqrt{C + x^2} \, dx$.

Note that this time, we're evaluating an indefinite integral. However, the rule still applies:

$$\int f(g(x)) g'(x) \, dx = \int f(u) \, du \text{ for } u = g(x).$$

Here, our substitution is a little subtle. We will use $u = C + x^2$, $du = 2x \, dx$, and $(u - C)^2 = x^4$. Our integral now becomes

$$\int x^5 \sqrt{C + x^2} \, dx = \frac{1}{2} \int (u - C)^2 \sqrt{u} \, du.$$

This can be evaluated with the power rule for integrals.

Integrals of symmetric functions

Noticing when your integrand is a symmetric function can come in handy, perhaps saving you lots of time when integrating. Let's briefly review what symmetric functions are.

An **even function** is any function that respects the property $f(x) = f(-x)$, such as $f(x) = x^2$.

An **odd function** is any function that respects the property $f(x) = -f(-x)$, such as $f(x) = \sin x$.

The integrals of such functions is as follows:

$$\text{For even functions, } \int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx. \text{ For odd functions, } \int_{-a}^a f(x) \, dx = 0.$$

Integration bound flipping, net, and total change over an interval I

Some integrals will give you an easier time if you evaluate them after flipping their bounds. However, we must be careful when doing so, because we can't just flip them freely. *Always* add a negative sign if you do so. Why? Letting $\frac{d}{dx}F(x) = f(x)$,

$$\begin{aligned}\int_a^b f(x) \, dx &= F(b) - F(a) \\ &= -(F(a) - F(b)) \\ &= -\int_b^a f(x) \, dx.\end{aligned}$$

Integration bound flipping can also be useful when investigating the **total change** of a function over an interval. This is one of two types of changes we can evaluate, the other being net change.

Net change is not too bad. For a differentiable function $F(x)$, the net change over $I = [a, b]$ is

$$\int_a^b F'(x) \, dx = F(b) - F(a).$$

The total change of $F(x)$ is slightly different, represented by $\int_a^b |F'(x)| \, dx$. We can look at this as a sum:

$$\int_a^b |F'(x)| \, dx = \sum_i \int_{x_i}^{x_{i+1}} F'(x) \, dx + \sum_j \int_{x_{j+1}}^{x_j} F'(x) \, dx,$$

$$\text{where } [x_i, x_{i+1}] \in I \mid [f(x_i), f(x_{i+1})] > 0, \text{ and } [x_j, x_{j+1}] \in I \mid [f(x_j), f(x_{j+1})] < 0.$$

Notice how all the bounds for j are flipped to make the integrals positive.

Chapter 5

Applications and Techniques of Integration

In the last chapter, we briefly introduced what an integral was. We will now dive deeper into the more technical details of integrals and how you can tackle specific forms of integrals.

5.1 Area between curves

We may first want to look at finding the area between two curves, $f(x)$ and $g(x)$. Previously, we only investigated the area between $f(x)$ and the x axis. There are a few ways to find the area between two curves, depending on what we mean by “area between curves”.

If $f(x) \geq g(x) \forall x \in I, I = [a, b]$, the integral is straightforward:

$$A_{f(x),g(x)} = \int_a^b (f(x) - g(x)) \, dx.$$

However, if $f(x)$ is not always larger than $g(x)$, and we'd like to find the *total area* between the curves, the integral becomes a bit more complicated. It's still not too bad, though:

$$A_{|f(x),g(x)|} = \sum_i \int_{x_i}^{x_{i+1}} (f(x) - g(x)) \, dx + \sum_j \int_{x_j}^{x_{j+1}} (g(x) - f(x)) \, dx,$$

where $[x_i, x_{i+1}] \in I \implies f(x) > g(x)$, and $[x_j, x_{j+1}] \in I \implies g(x) > f(x)$.

5.2 Volumes

While single variable integration primarily involves finding areas and consequences that arise from it, we can also use it to calculate volumes without having to use multivariable calculus. We'll look at a few methods to use when we want to calculate certain volumes.

Disk method

Say we have a continuous, integrable function $f(x)$. We would like to rotate that function about the x axis to create a volume. Since our $f(x)$ remains constant as we perform the rotation, we end up creating a circle that has area $\pi f(x)^2$. As we add up all of these little areas along our bounds of integration, we actually end up with the volume of our solid of interest.

An example is shown below. We have rotated $f(x) = x^2$ around the x axis, giving us a solid:

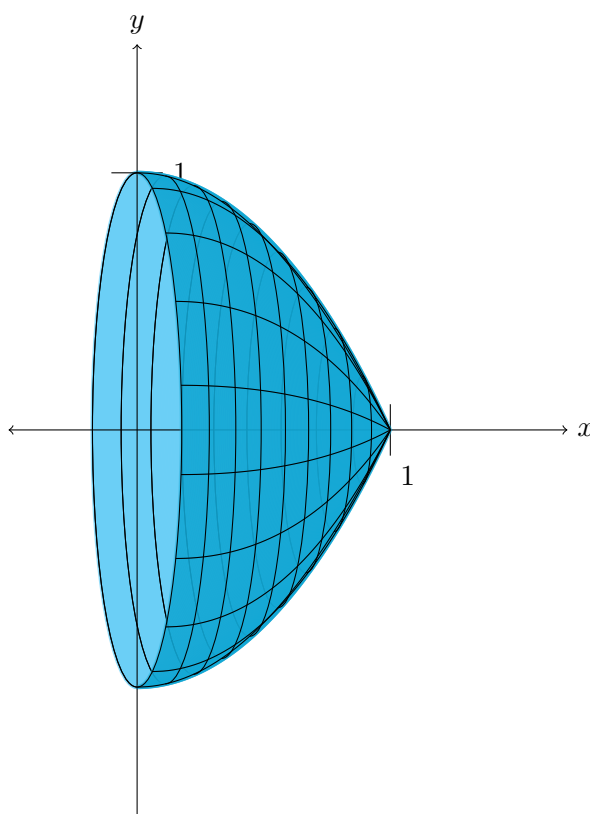


Figure 5.1: $f(x) = x^2$ rotated about the x axis.

Say we wanted to find the volume of this solid from $x = 0$ to $x = 1$. We would first find the area of any arbitrary circle in this region:

$$A = \pi f(x)^2.$$

We'll then add up all these areas to get our volume. This sounds an awful lot like taking the definite integral over our bounds. It is exactly that. Thus, the volume $\mathcal{V}$ of this solid is

$$\mathcal{V} = \int_0^1 A \, dx = \int_0^1 \pi f(x)^2 \, dx = \pi \int_0^1 x^4 \, dx = \frac{\pi}{5}.$$

This principle is known as the **disk method**.

Washer method

Sometimes, solids aren't as friendly as those we've explored above. For two functions $f(x)$ and $g(x)$ over an interval $I = [a, b]$, if $f(x) > g(x) \forall x \in I$, we could find the volume produced by rotating the *difference* of their functions around the x axis. For instance, take the functions $f(x) = \sin x$ and $g(x) = x^2$:

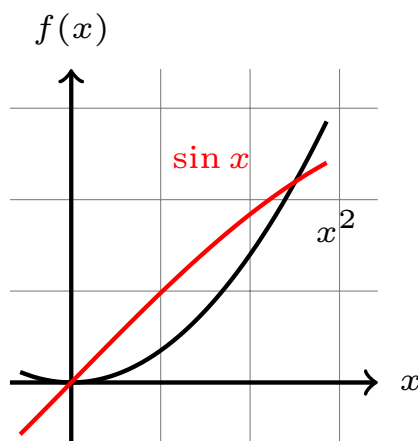


Figure 5.2: Our functions $f(x) = \sin x$ and $g(x) = x^2$.

It turns out that when we rotate the difference between these two functions around the x axis, the area of the circle formed is the *difference* between the area of the outer circle and the inner circle:

$$A = \pi (f(x)^2 - g(x)^2).$$

We can again sum up all the areas along our interval, yielding our volume.

$$\mathcal{V} = \pi \int_a^b (f(x)^2 - g(x)^2) \, dx.$$

Of course, if the interval contains points where $g(x) > f(x)$, we can still find the total volume. Just take each area as

$$A = \pi (g(x)^2 - f(x)^2).$$

This method of finding volume is known as the **Washer method**, because it looks like a washer.

The Shell method

So far, we have focused on solids created by rotating graphs about the x axis. What if it were instead rotated about the y axis? Let's take the function $f(x) = \cos(x^2)$ and rotate it about the y axis, generating a three dimensional solid. Below is an illustration of what looks like:

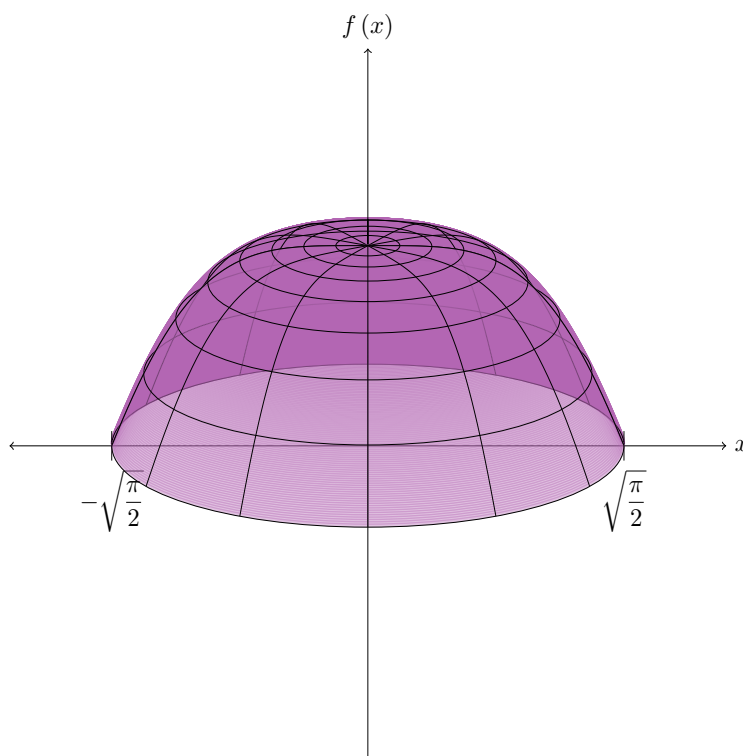


Figure 5.3: Rotating $f(x) = \cos(x^2)$ about the y -axis.

To find the volume of this solid, it's not as straightforward as the x axis rotation. We first investigate what it means to find the volume of a solid rotated about the y axis.

Because we are rotating about the y axis, integrating across our interval I means we expand radially outward with respect to the y axis. This means that every “differential” volume element we add on to our volume, we are basically adding a cylindrical washer with inner radius x and outer radius $x + dx$. A graph might help illustrate what we mean:

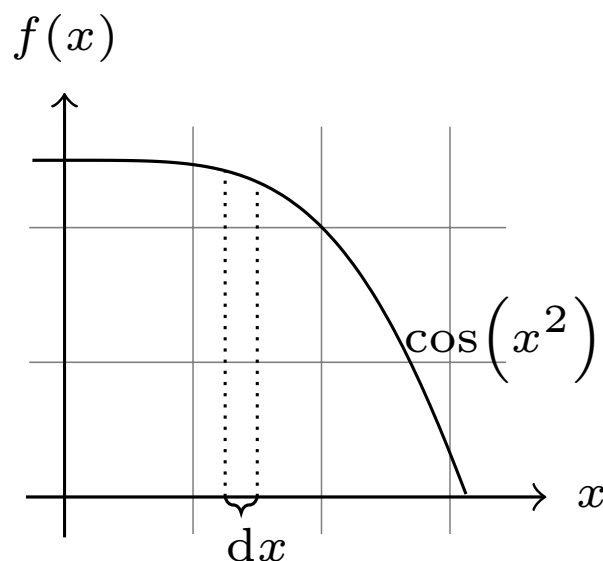


Figure 5.4: A visualization of each subsequent volume element being added onto the total volume of our solid. We take the volume of the outer cylinder and subtract it by the volume of the inner cylinder to get our volume element.

We first don't bother with the specifics of our function, and instead look more generally at the volume of a cylinder:

$$V = \pi r^2 h,$$

where r is just x and h is $f(x)$. For the i th cylinder, its volume must be

$$V_i = \pi r_i^2 h_i,$$

and for the $(i - 1)$ th cylinder,

$$V_{i-1} = \pi r_{i-1}^2 h_{i-1}.$$

Note that because the difference between r_i and r_{i-1} is small (and will tend toward 0 as we apply calculus), we can assume $h_i = h_{i-1}$. Their difference, which is our volume element

we wish to sum over, is thus

$$dV = V_i - V_{i-1} = \pi (r_i^2 h_i - r_{i-1}^2 h_{i-1}) = \pi h_i (r_i^2 - r_{i-1}^2).$$

As we push $i \rightarrow \infty$, we will converge at the actual volume of our solid rotated about the y axis. Writing it out mathematically tells us

$$\begin{aligned} V &= \sum dV \\ &= \lim_{n \rightarrow \infty} \pi \sum_{i=1}^n h_i (r_i^2 - r_{i-1}^2) \\ &= \lim_{n \rightarrow \infty} \pi \sum_{i=1}^n h_i (r_i + r_{i-1}) (r_i - r_{i-1}) \end{aligned}$$

Because r_i is just the x coordinate of our function, it makes sense to use the following substitutions:

$$r_i = a + \frac{k(b-a)}{n}, h_i = f(r_i), k \in [1, n].$$

Now we can combine everything.

$$\begin{aligned} V &= \lim_{n \rightarrow \infty} \pi \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \left(a + \frac{k(b-a)}{n} + a + \frac{(k-1)(b-a)}{n}\right) \\ &\quad \left(a + \frac{k(b-a)}{n} - \left(a + \frac{(k-1)(b-a)}{n}\right)\right) \\ &= \pi \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \left(2\left(a + \frac{k(b-a)}{n}\right) - \frac{b-a}{n}\right) \left(\frac{b-a}{n}\right) \\ &= 2\pi \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \left(a + \frac{k(b-a)}{n} - \frac{b-a}{2n}\right) \left(\frac{b-a}{n}\right) \\ &= 2\pi \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \left(\left(a + \frac{k(b-a)}{n}\right) \left(\frac{b-a}{n}\right) - \frac{(b-a)^2}{2n^2}\right). \end{aligned}$$

We see that as $n \rightarrow \infty$, $\frac{(b-a)^2}{2n^2}$ quickly becomes negligible. If we let

$$x = a + \frac{k(b-a)}{n},$$

you'll notice we simply have a Riemann sum which collapses into a definite integral:

$$V = 2\pi \int_a^b x f(x) \, dx.$$

This is known as the **Shell method**. If we go back to our original example of $\cos(x^2)$, we find that the volume of the solid is

$$V = 2\pi \int_0^{\sqrt{\frac{\pi}{2}}} x \cos(x^2) \, dx = \pi.$$

Pappus' first theorem

While this isn't exactly something to do with volume, it's still somewhat related (Pappus' second theorem deals more with volume, but that's beyond the scope of single variable calculus).

We imagine a path $\mathbf{c} : t \mapsto x$ parametrized by $t, t \in I, I = [a, b]$. Now, we rotate this path around the y axis to form a surface. We want to find out what the surface's area is. If we think about it, we can express a differential area element dA as the product of the circumference of rotating a point in $\mathbf{c} = x(t)$ around the y axis and the arc length differential $d\ell(\mathbf{c})$. However, this differential is just the product between the "speed" of the parametrized curve, $\|\mathbf{c}'(t)\|$ and dt . Thus,

$$dA = 2\pi x(t) \|\mathbf{c}'(t)\| \, dt.$$

To get the entire surface area, we just integrate:

$$A = \int_{\mathbf{c}} 2\pi x \, d\ell(\mathbf{c}) = \int_a^b 2\pi x(t) \|\mathbf{c}'(t)\| \, dt.$$

However, this is not a very pretty result. We will draw some other conclusions to make this result look better. If we recall the average value formula we looked at in Chapter 4, we can rewrite it in a neat way:

$$\bar{x} = \frac{1}{\ell(\mathbf{c})} \int_a^b x(t) \|\mathbf{c}'(t)\| \, dt.$$

The denominator is the arc length of our path $\mathbf{c}$. While we haven't covered arc length yet, this particular arc length is pretty straightforward: we just sum all the arc length

differentials.

$$\ell(\mathbf{c}) = \int d\ell(\mathbf{c}) = \int_a^b \|\mathbf{c}'(t)\| dt.$$

Using these two equations, we find that

$$\bar{x} \ell(\mathbf{c}) = \int_a^b x(t) \|\mathbf{c}'(t)\| dt = \frac{A}{2\pi}.$$

Finally, we arrive at our desired equation:

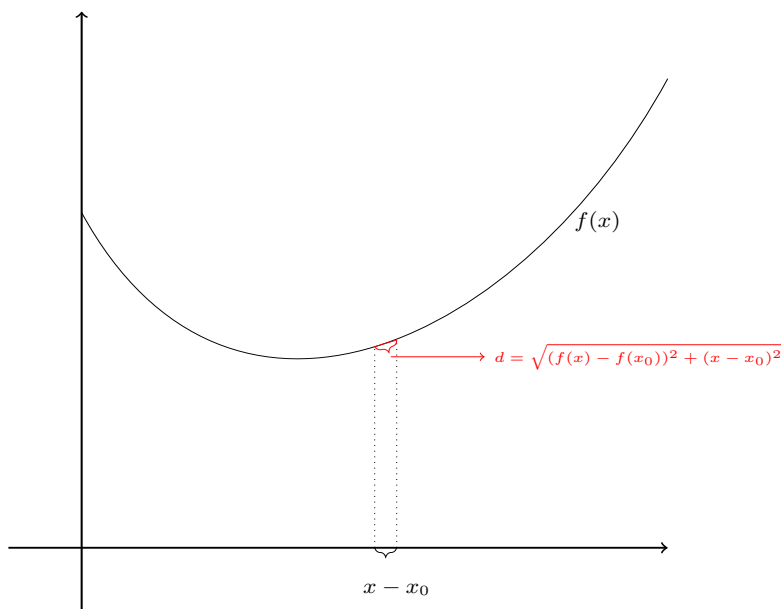
$$\boxed{A = 2\pi \bar{x} \ell(\mathbf{c})}$$

This equation just tells us that the surface area of *any* curve $\mathbf{c}$ rotated about the y axis is equal to the length of the curve multiplied by the average x value's distance revolved.

5.3 Arc length

Integration can not only be used to find volumes of unconventional solids, but are also useful in finding lengths of curves that are impossible or far more tedious to approach using noncalculus methods.

Let us have some interval $I = [a, b]$ and a continuous, differentiable function $f(x)$. We split it into n subintervals and find the distance between each subinterval. This is just the line connecting $f(x) - f(x_0)$ and $x - x_0$.



We reformulate this as our summand for any arbitrary k th and $k+1$ th subinterval within I :

$$d = \sqrt{\left(a + \frac{(k+1)(b-a)}{n} - \left[a + \frac{k(b-a)}{n}\right]\right)^2 + \left(f\left(a + \frac{(k+1)(b-a)}{n}\right) - f\left(a + \frac{k(b-a)}{n}\right)\right)^2},$$

$$k = 0, 1, \dots, n.$$

We can sum this expression over all k from 0 to n , which, as we take n to increase without bound, will give us a formula for arc length s :

$$s = \lim_{n \rightarrow \infty} \sum_{k=0}^n \sqrt{\left(a + \frac{(k+1)(b-a)}{n} - \left[a + \frac{k(b-a)}{n}\right]\right)^2 + \left(f\left(a + \frac{(k+1)(b-a)}{n}\right) - f\left(a + \frac{k(b-a)}{n}\right)\right)^2}.$$

We can simplify the above expression for s to get

$$s = \lim_{n \rightarrow \infty} \sum_{k=0}^n \sqrt{\left(\frac{b-a}{n}\right)^2 + \left(f\left(a + \frac{(k+1)(b-a)}{n}\right) - f\left(a + \frac{k(b-a)}{n}\right)\right)^2}.$$

Since each subinterval's length is $\Delta x = \frac{b-a}{n}$, $x = a + \frac{k(b-a)}{n}$, and $\lim_{n \rightarrow \infty} \frac{b-a}{n} = dx$, we can rewrite the equation accordingly:

$$\begin{aligned} s &= \lim_{n \rightarrow \infty} \sum_{k=0}^n \sqrt{\left(\frac{b-a}{n}\right)^2 + \left(f\left(a + \frac{(k+1)(b-a)}{n}\right) - f\left(a + \frac{k(b-a)}{n}\right)\right)^2} \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^n \sqrt{\Delta x^2 + (f(x + \Delta x) - f(x))^2} \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^n \sqrt{1 + \left(\frac{f(x + \Delta x) - f(x)}{\Delta x}\right)^2} \Delta x \\ &= \int_a^b \sqrt{1 + f'(x)^2} dx. \end{aligned}$$

This is the formula for arc length.

5.4 Techniques of integration

Here, we actually dive into some techniques to integration functions that aren't easily integrable by other methods presented earlier in the guide.

Integration by parts

In differentiation, we had the product rule

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x).$$

By integrating, we get

$$f(x)g(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx.$$

You can rearrange this function as needed. When most other methods fail, this is the one you want to turn to, but there's a chance you may get stuck in an endless loop of integrating by parts over and over again (it is a painful realization). Note that because we're using both $f'(x)$ and $g'(x)$, the parent functions must both be differentiable.

Example: Evaluate $\int xe^x dx$.

We will rearrange the equation such that

$$\int f(x)g'(x) dx = f(x)g(x) - \int f'(x)g(x) dx.$$

Let $f(x) = x, g'(x) = e^x$. Then

$$\begin{aligned} \int xe^x dx &= xe^x - \int e^x dx \\ &= e^x(x - 1). \end{aligned}$$

Example: Evaluate $\int \ln x dx$.

It may not seem like integration by parts is the best choice here, but if we arrange the equation as above and let $f(x) = \ln x, g(x) = 1$, we get

$$\begin{aligned} \int \ln x dx &= x \ln x - \int \frac{x}{x} dx \\ &= x(\ln x - 1) \end{aligned}$$

Partial fraction decomposition

Any rational function can be expressed as the quotient of two polynomials. If we would like to express the integral of our rational function, with numerator $P(x)$ and denominator $Q(x)$, we can write the following:

$$I = \int \frac{P(x)}{Q(x)} dx.$$

This seems incredibly annoying (and perhaps, on first glance, impossible) to evaluate! However, we can break down the rational function. Regardless of whether the degree of the denominator is larger or less than that of the numerator, we aim to reduce (or divide) the polynomial into a form

$$\frac{P(x)}{Q(x)} = \mathfrak{P}(x) + R(x),$$

where $\mathfrak{P}(x)$ consists of terms before the remainder term of the quotient, and $R(x)$ are the terms in the remainder (we use “remainder” here because we do polynomial division). $\mathfrak{P}(x)$ is easy to integrate; it’s just a polynomial and we can therefore use the integral power rule. However, $R(x)$ is a different story.

Review of partial fraction decomposition

For polynomials $p(x), q(x)$,

$$R(x) = \frac{p(x)}{q(x)} = \sum_i \frac{A_i}{\mathfrak{I}_i(x)},$$

where $\mathfrak{I}_i(x)$ are irreducible polynomials, and $A_i \in \mathbb{R}$. We then set up a set of equations in the form

$$\sum_i \frac{A_i}{\mathfrak{I}_i(x)} = \sum_i \frac{A_i \prod_{j \neq i} \mathfrak{I}_j(x)}{q(x)} \implies p(x) = \sum_i A_i \prod_{j \neq i} \mathfrak{I}_j(x).$$

From here, we can just match each term on the right hand side to the left, and solve for A_i . Below is an example.

Example: Decompose $\frac{1}{x^2 - 5x + 6}$.

$$\begin{aligned}
\text{Let } \frac{1}{x^2 - 5x + 6} &= \frac{A}{x - 2} + \frac{B}{x - 3} \\
&= \frac{A(x - 3)}{x^2 - 5x + 6} + \frac{B(x - 2)}{x^2 - 5x + 6} \\
\implies 1 &= A(x - 3) + B(x - 2) \\
\implies Ax &= -Bx, A = -B \\
\implies 1 &= -3A + 2A, A = -1, B = 1 \\
\text{Hence, } \frac{1}{x^2 - 5x + 6} &= \frac{1}{x - 3} - \frac{1}{x - 2}.
\end{aligned}$$

Now if we wanted to integrate, we could do so easily:

$$\int \frac{1}{x^2 - 5x + 6} dx = \int \left(\frac{1}{x - 3} - \frac{1}{x - 2} \right) dx = \ln|x - 3| - \ln|x - 2| + C.$$

Clever trigonometric identities and substitutions

Using trigonometric identities can sometimes be extremely useful, particularly when evaluating problems that have weird square roots or terms that you may have encountered while learning trigonometry (the applications are more ubiquitous than you may think). But first, let's look at a few important identities:

The classic	$\sin^2 x + \cos^2 x = 1$
Sine double identity	$\sin(2x) = 2 \sin x \cos x$
Cosine double identity	$\cos(2x) = \cos^2 x - \sin^2 x$
Sine half angle	$\sin^2 x = \frac{1 - \cos(2x)}{2}$
Cosine half angle	$\cos^2 x = \frac{1 + \cos(2x)}{2}$
Tangents and secants	$1 + \tan^2 x = \sec^2 x$

Also, keep in mind each trig function's derivative. They will come *very* handy here. The following is essentially an instruction manual on how to tackle all different types of trigonometric integrals.

Evaluating $\int \sin^m x \cos^n x dx$

Let $m = 2k + 1, n = 2k, k \in \mathbb{Z}$. Then

$$\begin{aligned}
\int \sin^m x \cos^n x dx &= \int (\sin^2 x)^k \cos^n x \sin x dx \\
&= \int (1 - \cos^2 x)^k \cos^n x \sin x dx.
\end{aligned}$$

Now, expand and substitute $u = \cos x$. The integral becomes much easier.

When m is even and n is odd, we do virtually the same thing except substitute $u = \cos x$ instead. If they're both even, we can use half angle identities repeatedly.

Evaluating $\int \sec^m x \tan^n x \, dx$

When m is even, meaning $m = 2k, k \in \mathbb{Z}$, we can rearrange the integral as

$$\begin{aligned} \int \sec^{2k} x \tan^n x \, dx &= \int (\sec^2 x)^{k-1} \tan^n x \sec^2 x \, dx \\ &= \int (1 + \tan^2 x)^{k-1} \tan^n x \sec^2 x \, dx. \end{aligned}$$

We can use $u = \tan x$ to simplify our integral.

When n is odd, that $n = 2k + 1, k \in \mathbb{Z}$, we make the following arrangement:

$$\begin{aligned} \int \sec^m x \tan^{2k+1} x \, dx &= \int \sec^{m-1} x \tan^{2k} x \sec x \tan x \, dx \\ &= \int \sec^{m-1} x (\sec^2 x - 1)^k \sec x \tan x \, dx \end{aligned}$$

Now we can use the substitution $u = \sec x$ to simplify our integral.

There is another case: evaluating $\int \tan^k x$. Here, we just repeatedly use the tangent and secant identity:

$$\int \tan^k x \, dx = \int \tan^{k-2} x \sec^2 x \, dx - \int \tan^{k-4} x \sec^2 x \, dx + \dots$$

We do this so that we can use the substitution $u = \tan x$ to simplify our integral.

5.5 Improper integrals

We say an integral is *improper* if it contains either ∞ or $-\infty$ as one of its bounds. In these cases, we define the improper integral to be

$$\int_a^\infty f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx.$$

These integrals are improper because it is mathematically illegal to plug in “ ∞ ” when evaluating points, however complacent physicists may be with abbreviating this step. We assume, of course, that $f(x)$ must be continuous and integrable along $[a, \infty)$. The same goes for having $-\infty$ as the lower bound of integration:

$$\int_{-\infty}^b f(x) \, dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) \, dx.$$

If f is continuous over all reals, there are a couple of ways to write its integral over $\mathbb{R}$. The first way is to write it as a sum

$$\int_{-\infty}^\infty f(x) \, dx = \int_{-\infty}^{a_1} f(x) \, dx + \sum_{i=1}^{n-1} \int_{a_i}^{a_{i+1}} f(x) \, dx + \int_{a_n}^\infty f(x) \, dx.$$

You can also write it as the limit

$$\int_{-\infty}^\infty f(x) \, dx = \int_{\mathbb{R}} f(x) \, dx = \lim_{a \rightarrow -\infty} \left(\lim_{b \rightarrow \infty} \left(\int_a^b f(x) \, dx \right) \right).$$

Chapter 6

Differential equations

6.1 What are they?

Differential equations, as their name may suggest, are equations involving both a function, $f(x)$, and its derivatives. Some examples include:

The pendulum equation: $\frac{d^2\theta}{dt^2} + \frac{g}{l} \sin \theta = 0$

Bessel's equation: $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - p^2)y = 0$.

Rate equation: $\frac{d\mathcal{A}}{dt} = -k\mathcal{A}$

When first working with differential equations, we'll be looking at *separable* differential equations. With the exception of the rate equation above, the other two examples are not separable. For instance, the pendulum equation is a nonlinear second order differential equation, while Bessel's equation is a linear second order differential equation.

Everything presented so far fall into a category called *ordinary differential equations*, where our function only takes one input. There are another class of differential equations called *partial differential equations*, which take in more than one input. We will not worry about those, but some examples include

Harmonic equations: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

Black-Scholes equation: $\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - r V = 0$.

Klein-Gordon equation: $\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2 c^2}{\hbar^2} \right) \psi(t, \mathbf{x}) = 0$

6.2 Separable differential equations

These equations are the first type of differential equations you will encounter because they are the least painful to solve.

The general form of such an equation is

$$\frac{dy}{dx} = f(x) g(y).$$

We can rearrange the equation and integrate:

$$\frac{dy}{g(y)} = f(x) dx \implies \int \frac{dy}{g(y)} = \int f(x) dx.$$

Let $G(y) + C$ be the integral of the left hand side and $F(x) + D$ be the integral of the right hand side. Our final solution of the differential equation looks like

$$G(y) = F(x) + \kappa, \text{ where } \kappa = D - C.$$

We notice, though, that κ isn't particularly well-defined. This is not a problem. Solutions to differential equations are families of functions that satisfy the differential equation. However, if we wish to determine one solution to our differential equation, we must have *initial conditions*, values for x and y that leave us with κ as the only unknown variable.

All solutions to a differential equation can be modeled by something called a *slope field*. An example for $\frac{dy}{dx} = x - y$ is shown below.

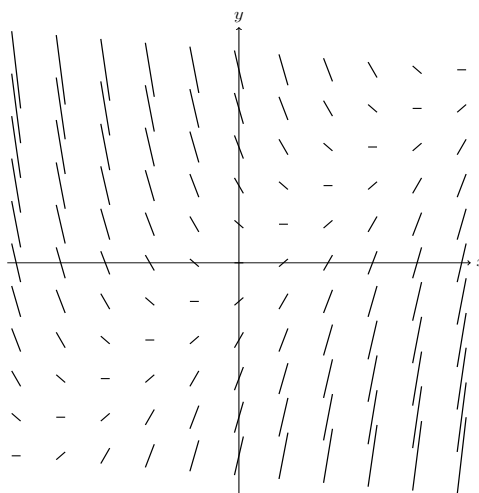


Figure 6.1: The slope field for $\frac{dy}{dx} = x - y$. A unique solution can be drawn by tracing your pen along the slope field.

6.3 Population growth models

When we first investigate differential equations, there are two population models we are most concerned with.

The first, called the **exponential growth model**, assumes a population has infinite resources to flourish in. This population, which we'll call P , and some constant that moderates its growth, c , can be represented by the differential equation

$$\frac{dP}{dt} = cP \quad (6.1)$$

Supposing we have an initial population P_0 at initial time t_0 , we can find the population P at any arbitrary time t by solving equation (6.1) to get

$$P = P_0 e^{c(t-t_0)} \quad (6.2)$$

Unfortunately, we live in a reality that does not provide infinite resources. If it did, we wouldn't need to be so worried about things like global warming or clean water (of course, there are still people who believe otherwise). A model that better represents our situation is the **logistic growth model**, which assumes a finite number of resources and a maximum sustainable population called the *carrying capacity*. Here, we use k to represent it. With these limiting factors, our differential equation becomes a little more complex:

$$\frac{dP}{dt} = cP \left(1 - \frac{P}{k}\right) \quad (6.3)$$

Notice that as our population nears k , the rate at which the population grows or shrinks approaches zero, which means our population levels off at k . If we are again given an initial population P_0 and initial time t_0 , we can solve for the population at any time t , this time yielding something slightly more intimidating.

$$P = \frac{k}{1 + \left(\frac{k-P_0}{P_0}\right) e^{-c(t-t_0)}} \quad (6.4)$$

6.4 Euler's method of approximation

While we can solve quite a range of differential equations just by the method of separation, there are many more we cannot solve without taking a course or reading a textbook on differential equations themselves (yeah, it's a pretty big topic). Some differential equations don't even have an analytic solution, and we must resort to computational methods instead. One of the ways to do this is known as **Euler's method of approximation**, and is widely used in computers to graph or solve equations. In fact, if you have a Mac and open up the Grapher app, go to Equation $\rightarrow$ Integration $\dots \rightarrow$ Method, there is an option to select "Euler", which uses this method of approximation.

To actually use this method, we first take a differential equation

$$\frac{dy}{dx} = F(x, y) \text{ with initial parameters } (x_0, y_0).$$

Because we've defined our parameters, this limits us to one solution $g(y) = f(x)$. We may now want to approximate a value $f(c)$ some distance away from x_0 . To do so, we first break the distance, $c - x_0$, into uniform intervals with length h . We then perform the recursion

$$x_{n+1} = x_n + h, y_{n+1} = y_n + F(x_n, y_n)$$

until we arrive at $x = c$. If we use larger step sizes, our error for $f(c)$ increases. This is illustrated below.

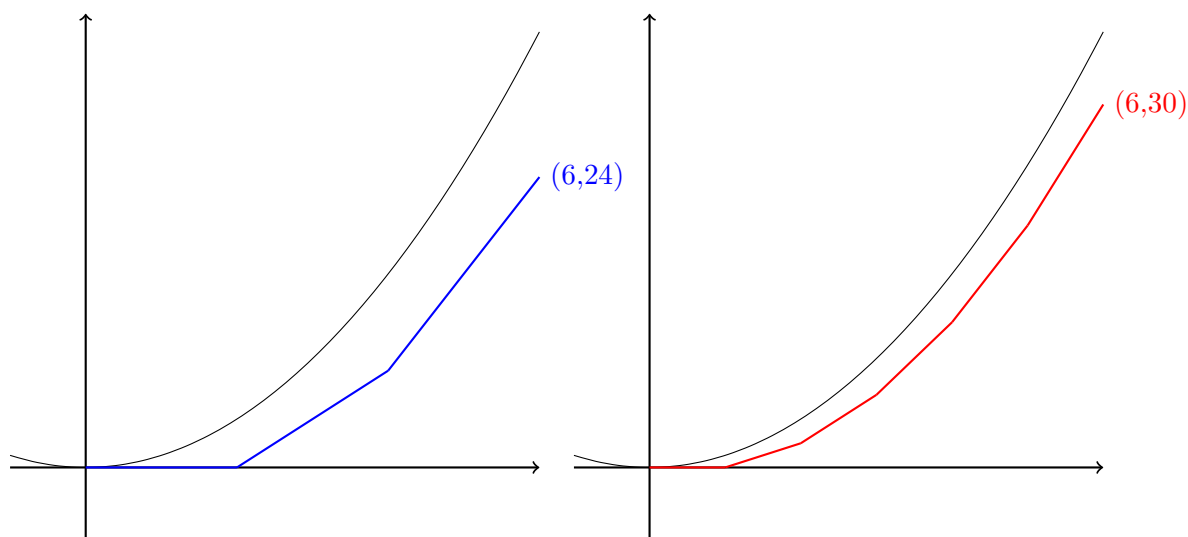


Figure 6.2: Euler's method for two different step sizes approximating $f(6)$ for $f(x) = x^2$.

Note how in the figure, our red approximation had a smaller step size and was thus more accurate. As our step sizes approach zero, Euler's method will become increasingly accurate. We will look at another example with an actual differential equation.

Example: For

$$\frac{dy}{dx} = 1 - y, \text{ and } f(1) = 0.$$

Start at $x = 1$ and approximate $f(0)$ using step size -0.5 .

We start with the information and construct the recursion

$$x_1 = 1 - 0.5, y_1 = 0 - 0.5(1) = -0.5.$$

We then take these values to find x_2, y_2 :

$$x_2 = 0, y_2 = -0.5 - 0.5 \left(\frac{3}{2} \right) = -0.5 - 0.75 = -\frac{5}{4}.$$

Using this step size, our approximation for $f(0) = -\frac{5}{4}$.

Chapter 7

Sequences and series

Sequences and series are something you've probably looked at a few times when taking algebra. However, without calculus, it's much harder to analyze sequences and series, and some of their properties are impossible to deduce without calculus.

7.1 Convergence

Here, we look at a few methods to determine whether a sequence or series converges, which is an important property.

Sequences

A sequence a converges if

$$\lim_{n \rightarrow \infty} \{a_n\} = L, L \in \mathbb{R}.$$

However, this definition is somewhat trivial and kind of useless. We still don't know how to rigorously tell whether a sequence converges. A more rigorous definition tells us that

$$\text{If } \forall \epsilon > 0, \exists M \in \mathbb{N} \mid |a_n - L| < \epsilon \forall n > M, \text{ then } \{a\} \text{ converges..}$$

Rewriting the sequence as a function (what some may call an *analytic continuation*) if possible can also help a lot. This allows you to perform functional analysis and find the limit of said function, which, if existent, would automatically imply the convergence of the sequence.

Example: Show that the sequence $\{a_n\}$, given by

$$\{a_n\} = \left(1 + \frac{1}{n}\right)^n,$$

converges to e .

This question asks us to show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

We let L be the limit of our sequence. We can thankfully turn this into a function and apply some calculus to it:

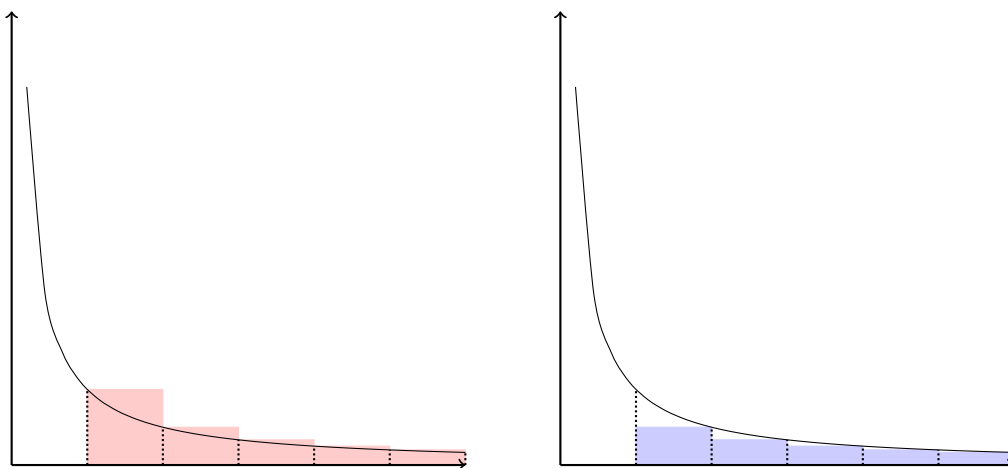
$$\begin{aligned} L &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \\ \ln L &= \lim_{x \rightarrow \infty} \frac{\ln\left(1 + \frac{1}{x}\right)}{\frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\frac{1}{1+\frac{1}{x}} \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} \quad (\text{L'Hôpital's rule}) \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} \\ &= 1 \\ \implies \ln L &= 1 \\ \implies L &= e. \end{aligned}$$

Series

There are numerous methods to determine whether a series converges. The first one you should always throw at a series is called the ***n*th term divergence test**. This asks whether the sequence that makes up the series converges. If not, don't waste your time! If the sequence doesn't even converge, the series will not either. If the series passes this test, there are a few other tests to determine convergence.

The **integral test** tells us that for eligible functions, we can rewrite the terms of the series as a function (like we did with sequences) and integrate the function. The function must satisfy three properties: positivity ($f(n) > 0$), monotonically decreasing ($f(n+1) < f(n)$), and continuous.

The series will follow the behavior of the integrated function. If the integral diverges, so does the series. The figure below aims to make sense of why this is.



The two figures above represent the left and right hand Riemann sums of the function $f(x) = \frac{1}{x}$, which is extrapolated from the series $S = 1 + \frac{1}{2} + \frac{1}{3} \dots$. Since the series starts at $n = 1$, we will integrate from 1 to ∞ . If the series started at another term, we would revise the integration bounds accordingly.

We see that the left figure is the left hand Riemann sum of $\frac{1}{x}$, and its area is always larger than that of the integral. The figure on the right is the right hand Riemann sum of $\frac{1}{x}$, and its area is always less than that of the integral. We focus on the left figure for now.

Since the area in red is greater than that of the integral, we can conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n} > \int_1^{\infty} \frac{1}{x} dx.$$

Evaluating our integral yields the expression

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{b \rightarrow \infty} \ln |b| - \ln 1,$$

meaning the integral diverges. This must also mean that the series diverges. Using the right figure to try and prove divergence will be futile, as all you can conclude is that the area of the righthanded figure is less than that of the integral. If the integral diverges, the series does not necessarily have to diverge. Therefore, it is inconclusive.

The right figure is more useful when we try to establish the convergence of a series. If, for instance, we had the series $\frac{1}{n^2}$, we could use the right handed Riemann sum. In this case,

$$\sum_{n=2}^{\infty} \frac{1}{n^2} < \int_1^{\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow \infty} -\frac{1}{b} + \frac{1}{1} = 1.$$

Notice, though, that we haven't included the first term of the series. That won't matter in our tests for convergence, because the first term is finite. So since the rest of the series is finite, we can conclude that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges.}$$

Another test, closely related to the integral test, is the ***p*-series test**. This covers series of the form

$$S = \sum_{n=1}^{\infty} \frac{1}{n^p}$$

and states that whenever $p > 1$, S converges. We look at why this is.

Consider turning this problem into an integral test. The function we wish to integrate is $\frac{1}{x^p}$. Integrating,

$$\int_1^{\infty} \frac{1}{x^p} dx = \int_1^{\infty} x^{-p} dx = \lim_{b \rightarrow \infty} \left(\frac{x^{-p+1}}{1-p} \right) \Big|_1^b.$$

Whenever $p > 1$, $1 - p < 0$, meaning

$$\lim_{b \rightarrow \infty} \left(\frac{x^{-p+1}}{1-p} \right) \Big|_1^b = \frac{1}{p-1}, \text{ which converges.}$$

This test is really useful when $f(x)$ is easily integrable. However, that is not always the case. When the above tests can't be easily applied, we can look towards the **limit comparison test**. This tells us that

$$\text{Given series } S_a = \sum_{n=1}^{\infty} a_n \text{ and } S_b = \sum_{n=1}^{\infty} b_n, \text{ if } \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = N, N \in \mathbb{R},$$

then S_a, S_b both converge or diverge.

Example: Does $\sum_{n=1}^{\infty} \frac{n+5}{3n4^n}$ converge?

Let $S_a = \sum_{n=1}^{\infty} \frac{n+5}{3n4^n}$ and $S_b = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n$. Performing the limit comparison test,

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{n+5}{3n4^n}\right)}{\left(\frac{1}{4}\right)^n} = \lim_{n \rightarrow \infty} \frac{n+5}{3n} = \frac{1}{3}.$$

Since we identify S_b as a geometric series that converges, S_a must therefore also converge.

Sometimes, we may not find a suitable series to perform the limit comparison test. Here, we can resort to the **ratio test**. It says

If, for series $\sum_{n=1}^{\infty} a_n$, $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = N$, $N \in \mathbb{R}$, $\sum a_n$ converges if $|N| < 1$ and diverges if $|N| > 1$.

If $|N| = 1$, the ratio test is inconclusive.

Example: Does $\sum_{n=1}^{\infty} \frac{n^3}{n!}$ converge?

Using the ratio test, we can produce the quotient

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{(n+1)^3}{(n+1)!} \right)}{\left(\frac{n^3}{n!} \right)} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^3} = 0.$$

Since this quotient approaches zero, we can conclude that the series converges.

Alternating series

So far, we have been looking at series where all the terms are positive and monotonically decreasing. There are also alternating series, which have the form

$$\sum_{n=1}^{\infty} a_n, \text{ where } a_n = (-1)^n f(n).$$

There are a few ways to evaluate the convergence of alternating series. The first is called the **absolute convergence of alternating series**. The philosophy is that if $\sum |a_n|$ converges, $\sum a_n$ must converge too. Since some of the terms are negative, if the absolute value converges then it must mean that the regular series does too. However, $\sum a_n$ can still converge even if $\sum |a_n|$ doesn't.

If we cannot firmly establish convergence of $\sum a_n$ by the absolute convergence test, it can still converge if it satisfies two conditions:

$$\sum a_n \text{ converges if } \begin{cases} \mathbf{1.} |a_{n+1}| < |a_n| \text{ and} \\ \mathbf{2.} \lim_{n \rightarrow \infty} a_n = 0 \end{cases}$$

Example: Does $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converge?

Trying the absolute convergence test, we find that it does not converge as it becomes the harmonic series. However, we know that $\frac{1}{n+1} < \frac{1}{n}$, and $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. This series satisfies the other two conditions, so it converges. We say series that fail the absolute convergence test but satisfy the other two conditions converges *conditionally*.

7.2 Taylor series

Perhaps we have a function we'd like to do calculus on (in particular, integrate). There are some functions that are impossible to integrate using other methods we have introduced in this guide. For instance, $f(x) = e^{-x^2}$ can't be integrated with other methods we mentioned earlier.

In these cases, we can decompose the function into other functions that are easier to integrate. Here, we will use a sum of polynomials because, as we've come to see, they are the easiest to do calculus on. You can also decompose them into sums of sine and cosine waves, but we won't worry about that (they are known as *Fourier series*). This sum of polynomials is better known as a function's **Taylor series** or **Taylor expansion**.

These polynomials come in the form

$$T(x) = \sum_{n=0}^{\infty} c_n (x - a)^n,$$

and aim to converge to $f(x)$ around the point $x = a$.

The primary goal to decompose our function into polynomials means that our polynomials must converge to $f(x)$ as we add more terms to the polynomial. This implies identical behavior, which means they must have the same zeroth derivative, first derivative, and so on.

To have such behavior, our function will equal

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(a)(x - a)^n}{n!}$$

for our series centered around $x = a$. Here, $f^n(a)$ is the n th derivative of $f(x)$ at $x = a$. Take a few derivatives to see that it matches our goals for the polynomial decomposition.

Convergence of $T(x)$ to $f(x)$

While we aim to have a polynomial decomposition that converges to $f(x)$ for all values of x , this is not always the case. There are three possibilities of convergence:

$$T(x) \text{ converges to } f(x) \begin{cases} \text{only at } x = a. \\ \text{some "distance" around } a, \text{ known as the } \textit{radius of convergence}. \\ \forall x \in \mathbb{R}. \end{cases}$$

Finding the radius and interval of convergence

The radius and interval of convergence tells us where $T(x)$ is a valid approximation of $f(x)$. This is important because if we try integrating over values outside of the interval, we will run into some big problems.

We can actually find the radius and interval of convergence using some of the series tests presented earlier. To find the radius, we use the ratio test to determine at which points the ratio test becomes inconclusive. That is, when the ratio test is equal to 1. We do this because when the ratio is less than 1, the series converges, but when it's greater than 1, it diverges. Hence, when the ratio equals 1, we have our radius of convergence. Once we find the radius, we can substitute the endpoints into the series and evaluate to determine convergence. Below are a few worked examples.

Example: Find the radius and interval of convergence for $\sum_{n=0}^{\infty} n! x^n$.

When we use the ratio test, we try to find x such that the ratio test equals 1.

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| = 1$$

$$\lim_{n \rightarrow \infty} |(n+1)x| = 1.$$

Evaluating the limit gives us ∞ as $n \rightarrow \infty$, meaning there are no values of x outside $x = 0$ where the series converges. This means the series converges at $x = 0$ only and has radius of convergence 0.

Example: Find the radius and interval of convergence for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$.

Again, we use the ratio test.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2(n+1)}}{(2(n+1))!} \cdot \frac{(2n)!}{(-1)^n x^{2n}} \right| &= \lim_{n \rightarrow \infty} \left| \frac{x^2}{(2n+1)(2n+2)} \right| \\ &= 0 < 1. \end{aligned}$$

No matter what value of x we choose, our limit is always less than 1, meaning the series converges $\forall x \in \mathbb{R}$. This is actually the Taylor expansion of $\cos x$.

Example: Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n \left(\frac{2x-5}{3} \right)^n$.

Using the ratio test again, we find

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{(n+1) \left(\frac{2x-5}{3} \right)^{n+1}}{n \left(\frac{2x-5}{3} \right)^n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \cdot \left(\frac{2x-5}{3} \right) \right| \\ &= \left| \frac{2x-5}{3} \right| \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \right| \\ &= \left| \frac{2x-5}{3} \right|. \end{aligned}$$

To find the points of convergence, we just need to find points that satisfy $\left| \frac{2x-5}{3} \right| < 1$. We find that $1 < x < 4$ works, so we evaluate $x = 1, 4$ to determine the interval of convergence. As you might be able to tell, the radius of convergence is $\frac{3}{2}$.

At $x = 4$, we can simplify the series to

$$\sum_{n=0}^{\infty} n, \text{ which diverges.}$$

However, at $x = 1$, our series simplifies to

$$\sum_{n=0}^{\infty} (-1)^n n.$$

None of the alternating series tests indicate this series converges, so we conclude it must diverge. Hence, the interval of convergence for this series is $(-1, 4)$.

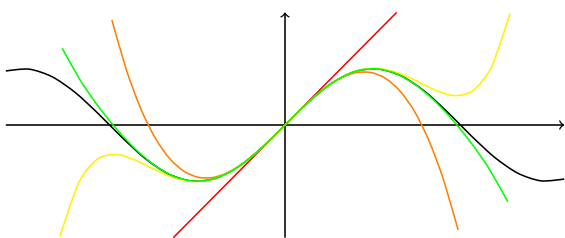
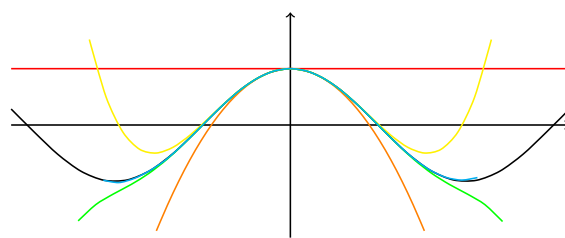
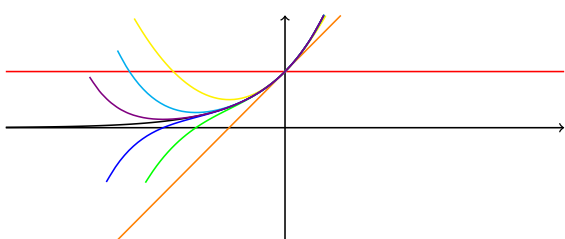
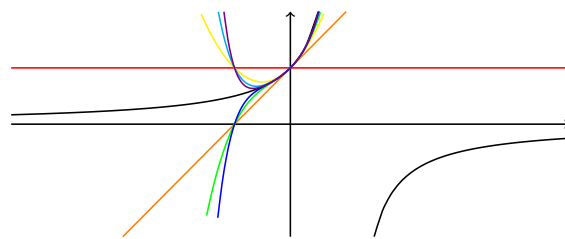
Important series and notes

There are four key Taylor series to take note of. The way they are written here are in a special form called *Maclaurin series*, which are just Taylor series centered around $x = 0$. Here they are:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Note that the Maclaurin expansion for $\frac{1}{1-x}$ has radius of convergence 1. For practice, you can find the interval of convergence. Below are figures showing each function and Taylor expansions of better accuracy.

Figure 7.1: $\sin x$ Figure 7.2: $\cos x$ Figure 7.3: e^x Figure 7.4: $\frac{1}{1-x}$

As the color goes from red to violet, the Taylor expansion becomes more accurate. You can see how each subsequent expansion better approximates the actual function. Also note how for $\frac{1}{1-x}$, the expansions all seem to diverge after a certain point. That is its radius of convergence.

For a Maclaurin series involving composite functions, you can actually do the following:

$$(f \circ g)(x) = \sum_{n=0}^{\infty} \frac{(f \circ g)^n(0)x^n}{n!} = \sum_{n=0}^{\infty} \frac{f^n(0)g(x)^n}{n!}.$$

For instance, we can easily generate the Maclaurin series for e^{-x^2} now: it's just

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!}.$$

Let's try extracting some information out of it that we otherwise couldn't do. Integrating even over a small range, from -3 to 3 should give us an answer very close to the integral over all the real numbers. After all, the graph looks like this:

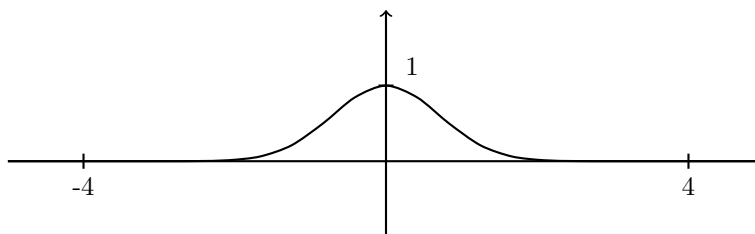


Figure 7.5: A graph of $f(x) = e^{-x^2}$.

Taking the first 30 nonzero terms, we integrate. Here are the three results side by side.

$$\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi} \quad \int_{-3}^3 e^{-x^2} dx = 1.77241469652 \quad \int_{-3}^3 \sum_{n=0}^{30} \frac{(-x^2)^n}{n!} dx = 1.77241816403.$$

The second integral deviates by 0.00221%, while the third deviates by 0.00201%. That's pretty close, and definitely a much better result than having no answer at all because we can't integrate the function.

Determining error

While Taylor polynomials are awesome, they are very time consuming, especially when you try to add up all the terms to obtain better approximations. Therefore, you may choose to abbreviate a few terms, which will save your time, but also create inaccuracies. Determining such errors are a good way to evaluate whether your approximation is still valid at certain points in the function's domain. If your error stays within a certain margin of your choice, then all is good.

For alternating series that adhere to the alternating series tests, this is somewhat straight-

forward. If we let the remainder, or error, be termed $R_n(x) = |f(x) - T_n(x)|$, where $f(x)$ is our function and $T_n(x)$ is the n th degree approximation of it,

$$R_n(x) \leq |T_{n+1}(x) - T_n(x)|.$$

In other words, the remainder is no larger than the $(n + 1)$ th term of the Taylor series.

For non-alternating polynomials, the error bound is a bit harder to understand. Each successive Taylor expansion better models $f(x)$ than its predecessor. $T_{n+1}(x)$ has the same behavior as $f(x)$ up through the $(n + 1)$ th derivative, while $T_n(x)$ only has the same behavior until the n th derivative. Therefore, we can say that $T_n(x)$ may not necessarily match the $(n+1)$ th derivative of $f(x)$, which introduces a potential source of error.

We thus wish to place an upper limit on the error bound. To do so, we must assume the maximum value $|f^{n+1}(x)|$ can take on between $x = a$ and the x value we'll use to evaluate error. Let's term this M . Then, for a Taylor series centered around $x = a$,

$$R_n(x) = |f(x) - T_n(x)| \leq \sup \left(\frac{|f^{n+1}(x)| |x - a|^{n+1}}{(n + 1)!} \right) = \frac{M |x - a|^{n+1}}{(n + 1)!}.$$

This is known as the **Lagrange error bound**.

Chapter 8

Parametric and polar functions

8.1 Parametric functions

As their name suggests, parametric functions are functions that depend on a parameter. In the xy plane, y is usually parametrized by x . However, with parametric functions, we generally have another variable, usually termed t . When evaluated over its domain, a range of x and y values are produced, which can allow us to draw curves on $\mathbb{R}^2$ (you'll know this as the xy plane) that functions of x itself cannot otherwise generate (they would violate the vertical line test). Some parametric functions can be reduced to a function of x itself or equations of x and y . Mathematically, our parametric map $\mathcal{P}$ does the following:

$$\left. \begin{array}{l} \mathcal{P} : \mathbb{R} \rightarrow \mathbb{R}^2 \\ \mathcal{P} : (t) \mapsto (x(t), y(t)) \end{array} \right\} \text{ via } \left\{ \begin{array}{l} x = x(t) \\ y = y(t) \end{array} \right.$$

There are other variations we can add to parametrize more objects and surfaces, but we're not going to worry about them in single variable calculus.

With these parametric functions, the properties we care most about them are their derivatives and arc length.

Derivatives of parametric functions

A quick note on convention: We denote time derivatives, such as $\frac{dx}{dt}$, with a dot on top of what we wish to differentiate. Thus, $\frac{dx}{dt} = \dot{x}$. Second time derivatives are denoted with two dots, and so on.

To gain some intuition on how to take derivatives of parametric functions, we take a look at a function $y = h(x)$, parametrized by $y = y(t)$ and $x = x(t)$. An image of such a

function is shown below.

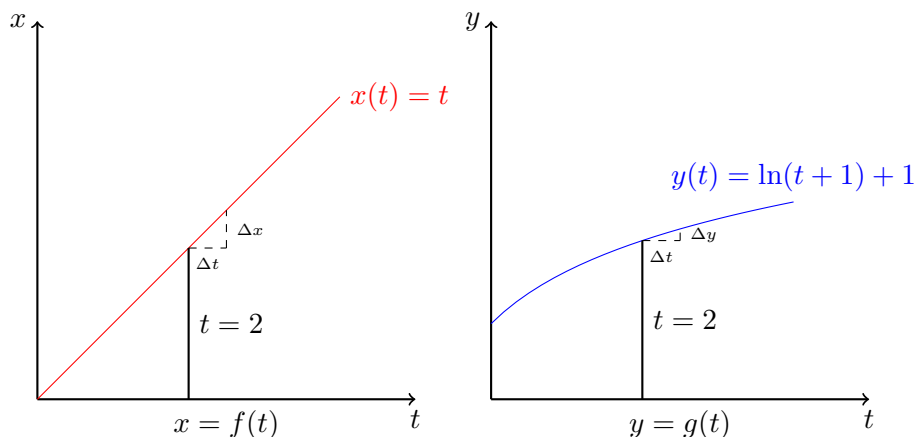


Figure 8.1: A parametric function mapped by $x = t, y = \ln(t + 1) + 1$.

Notice that for the same Δt , there is a corresponding change in Δx and Δy , not necessarily of the same magnitude. We can also represent Δx and Δy as

$$\Delta x = x(t + \Delta t) - x(t), \quad \Delta y = y(t + \Delta t) - y(t).$$

As we let $\Delta t \rightarrow 0$, we will better approximate the slope at these points, $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

$$\frac{dx}{dt} = \lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t}, \quad \frac{dy}{dt} = \lim_{\Delta t \rightarrow 0} \frac{y(t + \Delta t) - y(t)}{\Delta t}.$$

Since we're calculating Δy and Δx over the same Δt , we can also directly compare Δy and Δx as $\Delta t \rightarrow 0$. If we write it out,

$$\frac{dy}{dx} = \lim_{\Delta t \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta t \rightarrow 0} \frac{y(t + \Delta t) - y(t)}{x(t + \Delta t) - x(t)} = \frac{\left(\lim_{\Delta t \rightarrow 0} \frac{y(t + \Delta t) - y(t)}{\Delta t} \right)}{\left(\lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} \right)} = \frac{\dot{y}}{\dot{x}}.$$

We will now try to approach the second derivative. It's a lot less pleasing than the first derivative and there's not an intuitive way to think about it (not that I know of, anyway). What we would like to find is

$$\frac{d}{dx} \left(\frac{dy}{dx} \right).$$

Thankfully, we have already found the first derivative, and now must try to understand what it means to find another derivative with respect to x . For this, we'll use some algebra.

We work through some steps to arrive at the second derivative, some of which seem completely illegal.

$$\begin{aligned}\frac{d}{dx} \left(\frac{dy}{dx} \right) &= \frac{d}{dx} \frac{\dot{y}}{\dot{x}} \\ &= \frac{dt}{dx} \frac{d}{dt} \frac{\dot{y}}{\dot{x}} \\ &= \frac{\frac{d}{dt} \frac{\dot{y}}{\dot{x}}}{\frac{dx}{dt}}.\end{aligned}$$

Now we can actually take derivatives, since $\dot{x}$ and $\dot{y}$ are both functions of t . You can practice taking the derivatives here, but should end up with

$$\frac{d^2y}{dx^2} = \frac{\ddot{y}\dot{x} - \dot{y}\ddot{x}}{\dot{x}^3}(\text{ew.})$$

Example: Let $x(t) = \sqrt{t}$, $y(t) = \frac{1}{4}(t^2 - 4)$.

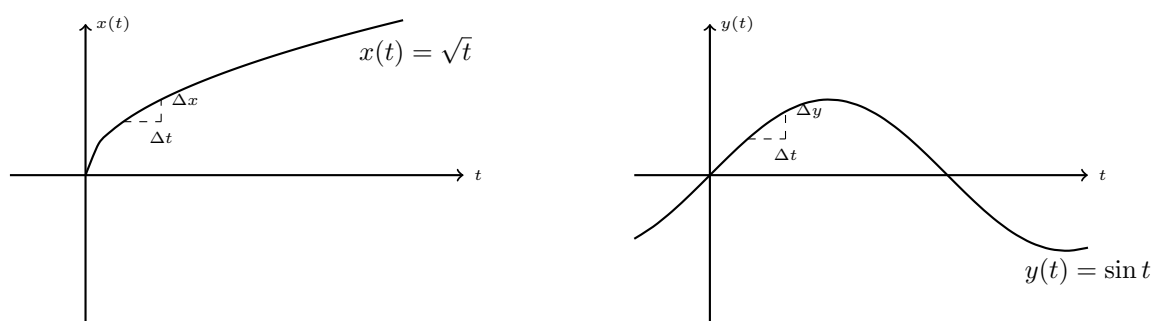
- a. For what values of x does the function exist?
- b. For what values of t is the function concave up?

We would like to see the range of $x(t)$ to determine values for which x exists. Since square roots cannot be negative, we conclude that x exists when $x \in [0, \infty)$.

To find where the function is concave up, we need to analyze the second derivative and find values where it is positive. Practice this, but you should end up with $3t$. Whenever $t > 0$, $3t > 0$, implying concave up. Hence, the function is concave up $\forall t \in (0, \infty)$.

Arc length in parametric functions

There are a few ways to think about the arc length of parametric functions. We first look at a visual method. Consider the parametrization $(x, y) = (\sqrt{t}, \sin t)$. Their graphs are shown below.



For every increment Δt , there is a corresponding Δx and Δy that is reflected in the xy plane. Let $\delta_x = (\Delta x)\mathbf{i}$ and $\delta_y = (\Delta y)\mathbf{j}$ be vectors, and let their vector sum be denoted $\delta_{x+y} = \delta_x + \delta_y$.

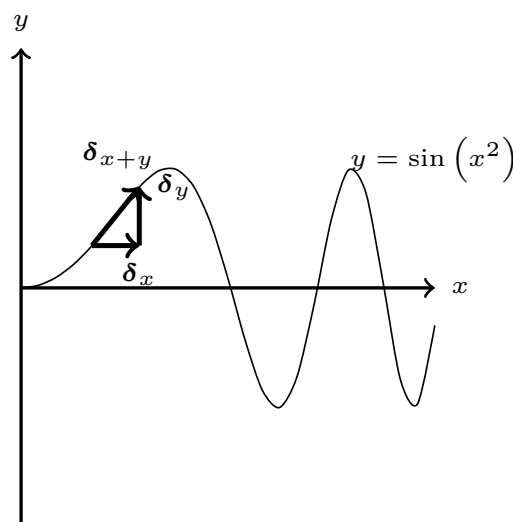


Figure 8.2: Our parametric figure in the xy plane.

We see that the vector sum is just a small piece of the curve. This means that if we sum up all the arc lengths as we bring Δx and Δy to zero, we can get the arc length. In other words, our arc length s is represented as

$$s = \lim_{\Delta t \rightarrow 0} \sum \|\delta_{x+y}\| = \lim_{\Delta t \rightarrow 0} \sum \frac{\|\delta_{x+y}\|}{\Delta t} \Delta t.$$

Writing our equation with variables given, we get that

$$s = \lim_{\Delta t \rightarrow 0} \sum \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2} \Delta t \quad (\text{since } \|\delta_{x+y}\| = \sqrt{\Delta x^2 + \Delta y^2}).$$

As we take this limit towards zero, our summation sign is replaced by an integral, giving

us the arc length formula.

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

An algebraic way to approach the formula:

We know the arc length formula for regular functions:

$$s = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

If we use a substitution $x = x(t)$, we find that $dx = x'(t)dt$. This means

$$\begin{aligned} \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx &= \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \frac{dx}{dt} dt \\ &= \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \left(\frac{dx}{dt}\right) dt \\ &= \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt. \end{aligned}$$

8.2 Polar functions

Polar functions are interesting and are included in the section on parametric functions because, in many ways, they are parametric functions themselves. Prior to learning calculus, you may have encountered problems where you had to solve for the intersection of two polar functions over a range of θ . While it may have seemed pointless at the time, we will soon discover that they find uses in calculus.

Areas in polar functions

By the nature of calculus, we can find the area of polar functions by first slicing the function into numerous snippets. As we decrease the value of θ , each snippet better approximates that of an isosceles triangle, with legs of length $f(\theta)$ and base $f(\theta)d\theta$. You can think of the base as an arbitrarily small arc spanned by a circle with radius $f(\theta)$ over the angle $d\theta$.

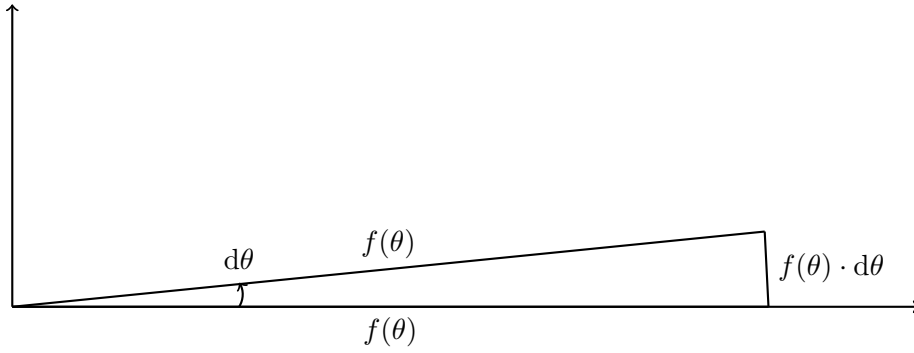


Figure 8.3: A little displacement in θ , $d\theta$, forms a triangle with base $f(\theta)d\theta$ and height $\approx f(\theta)$.

This illustration gives us a basic idea of what we are summing to find the integral. We can use the labels in the diagram to give us the area of a triangle.

$$\text{Area}_{\Delta} = \frac{1}{2} b h = \frac{1}{2} f(\theta) d\theta f(\theta) = \frac{1}{2} f(\theta)^2 d\theta.$$

Now we use the polar analogue of Riemann sums to get an approximation of our area we wish to determine. We assume we'd like to find the area of some polar function $f(\theta)$ over the bounds θ_1 and θ_2 .

$$A \approx \frac{1}{2} \sum_{k=0}^n f\left(\theta_1 + \frac{k(\theta_2 - \theta_1)}{n}\right)^2 \left(\frac{\theta_2 - \theta_1}{n}\right).$$

As we did earlier with Riemann sums, we now take $n \rightarrow \infty$ and get

$$A = \lim_{n \rightarrow \infty} \frac{1}{2} \sum_{k=0}^n f\left(\theta_1 + \frac{k(\theta_2 - \theta_1)}{n}\right)^2 \left(\frac{\theta_2 - \theta_1}{n}\right) = \boxed{\frac{1}{2} \int_{\theta_1}^{\theta_2} f(\theta)^2 d\theta}.$$

Derivatives in rectilinear and polar coordinates of polar functions

Recall the maps between polar and rectilinear coordinates:

$$\left[\begin{array}{l} \mathcal{R} : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \\ \mathcal{R} : (r, \theta) \mapsto (x, y) \end{array} \right\} \text{ via } \left\{ \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \end{array} \right\} \text{ and } \left[\begin{array}{l} \mathcal{P} : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \\ \mathcal{P} : (x, y) \mapsto (r, \theta) \end{array} \right\} \text{ via } \left\{ \begin{array}{l} r^2 = x^2 + y^2 \\ \theta = \tan^{-1}\left(\frac{y}{x}\right) \end{array} \right\}$$

We have all these variables to work with, and we may wish to find the derivative with respect to some of these variables. We'll discuss a few important derivatives.

We are interested in first finding $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$. Keeping in mind a few important things,

we see that they're not too bad:

$$\begin{aligned}\frac{dx}{d\theta} &= \frac{d}{d\theta}[r(\theta) \cos \theta] = \frac{dr}{d\theta} \cos \theta - r(\theta) \sin \theta \\ \frac{dy}{d\theta} &= \frac{d}{d\theta}[r(\theta) \sin \theta] = \frac{dr}{d\theta} \sin \theta + r(\theta) \cos \theta\end{aligned}$$

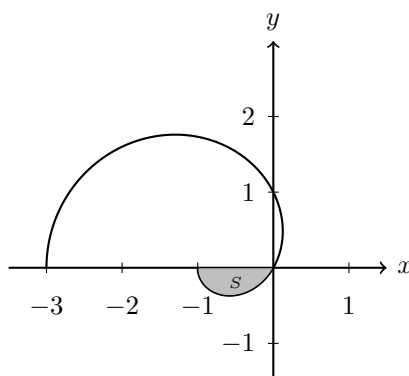
As we mentioned earlier, parametric and polar functions are quite similar. In parametric functions, we found that

$$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}}.$$

With polar functions, the mechanism is the same, with t replaced by θ .

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{\left(\frac{dr}{d\theta} \sin \theta + r(\theta) \cos \theta\right)}{\left(\frac{dr}{d\theta} \cos \theta - r(\theta) \sin \theta\right)}.$$

Example: The polar curve $r = 1 - 2 \cos \theta$ for $0 \leq \theta \leq \pi$ is shown below. Let S be the shaded region in the third quadrant bounded by the curve and the x -axis¹.



1. Write an integral expression for the area of S .
2. Write expressions for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ .
3. Write an equation in terms of x and y for the line tangent to the graph of the polar curve at the point where $\theta = \frac{\pi}{2}$. Show the computations that lead to your answer.

¹Taken from a previous Calculus BC examination

Recall our formula for the area:

$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} f(\theta)^2 d\theta.$$

$f(\theta)$ is given to us by r . We would like to find our bounds of integration, θ_1 and θ_2 . Do to so, we solve for when $\theta = -1, 0$. We find that $\theta = 0$ gives -1 . Solving for $\theta = 0$:

$$\begin{aligned} 0 &= 1 - 2 \cos \theta \\ 1 &= 2 \cos \theta \\ \theta &= \cos^{-1} \left(\frac{1}{2} \right). \end{aligned}$$

The only value of θ that solves the equation and falls in our range is $\theta = \frac{\pi}{3}$. Therefore, our integral expression for area is

$$A = \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 - 2 \cos \theta)^2 d\theta.$$

We now wish to find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ . Recall that

$$\frac{dx}{d\theta} = \frac{d}{d\theta} (r \cos \theta) \quad \text{and} \quad \frac{dy}{d\theta} = \frac{d}{d\theta} (r \sin \theta).$$

Now we just do some algebra.

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta} (1 - 2 \cos \theta) \cos \theta & \frac{dy}{d\theta} &= \frac{d}{d\theta} (1 - 2 \cos \theta) \sin \theta \\ &= \frac{d}{d\theta} (\cos \theta - 2 \cos^2 \theta) & &= \frac{d}{d\theta} (\sin \theta - \sin 2\theta) \\ &= \boxed{2 \sin \theta \cos \theta - \sin \theta} & &= \boxed{\cos \theta - 2 \cos 2\theta}. \end{aligned}$$

Finally, we wish to construct a tangent line to the curve at $\theta = \frac{\pi}{2}$. Recall the point-slope form of a line

$$y - y_1 = m(x - x_1).$$

Here, the point we wish to find is $(x, y) = (r \cos \theta, r \sin \theta)$ at $\theta = \frac{\pi}{2}$, and the slope of the tangent line is nothing but $\frac{dy}{dx}$. So, what we really wish to find is

$$y - r \sin \theta = \frac{dy}{dx} (x - r \cos \theta) \quad \text{at } \theta = \frac{\pi}{2}.$$

Finding (x, y) is not bad, it's just $(0, 1)$. To find $\frac{dy}{dx}$, however, we must divide $\frac{dy}{d\theta}$ by $\frac{dx}{d\theta}$. As such,

$$\begin{aligned}\frac{dy}{dx} &= \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} \\ &= \frac{\cos \theta - 2 \cos 2\theta}{\sin 2\theta - \sin \theta} \\ &= \frac{2}{-1} \\ &= \boxed{-2}.\end{aligned}$$

We combine all the information to get that the tangent line to the curve at $\theta = \frac{\pi}{2}$ is

$$\boxed{y - 1 = -2x}.$$

Arc length in polar coordinates

Just as we explored arc length in rectilinear coordinates and parametric functions, we will look at how to express it in polar coordinates. I briefly mentioned that polar functions are sort of like parametric functions in that θ is our parameter. Here, we really see the relevance of the statement. Recall our formula for arc length for parametric functions:

$$s = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Instead of using t as the parameter, we use θ and our polar representations of x and y . Rewriting the integral,

$$s = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta, \text{ where } \begin{cases} x = r(\theta) \cos \theta \\ y = r(\theta) \sin \theta \end{cases}$$

Using the product rule, we see

$$\begin{aligned}s &= \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\&= \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dr}{d\theta} \cos \theta - r \sin \theta\right)^2 + \left(\frac{dr}{d\theta} \sin \theta + r \cos \theta\right)^2} d\theta \\&= \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dr}{d\theta}\right)^2 (\sin^2 \theta + \cos^2 \theta) + r^2 (\sin^2 \theta + \cos^2 \theta)} d\theta \\&= \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.\end{aligned}$$

The end of this chapter marks most of what one will see in a first year course for calculus. It also covers most material that could be tested on the AP Calculus BC examination. Hopefully, you'll have gained insight into the world of calculus and its numerous applications in important aspects of life. While most take calculus to be the end of their mathematical voyage, it is, in reality, barely even the beginning. There are many topics that develop from calculus or use relevant concepts, including things like real analysis, differential geometry, or algebraic topology, and learning of such connections is nothing less than joyful.

Armed with this knowledge, you are ready to take on harder maths. Somewhere good to start is considering linear algebra or multivariable calculus.

Additional information, including the meaning of symbols I used throughout the text, along with proofs and derivations of some formulas we used, is included in the appendices below.

Appendix A

This appendix contains the meaning behind all the symbols I've used throughout the text, particularly those that have been seen by fewer people. I will write these symbols in the order that they show up throughout the text.

$\lim_{x \rightarrow a}$ – Limit as x approaches the point $x = a$.

$\pm, \mp$ – Plus or minus, minus or plus.

$\iff$ – If and only if.

$\forall$ – For all.

$\in$ – Is an element of.

$\exists$ – There exists.

$\mathbb{R}$ – The set of all real numbers.

$\circ$ – Of. A composition. For instance, $f \circ g$ means the composition of f on g .

$|$ (when used in between two statements) – such that. For instance, the conditional $P | Q$ says P such that Q , where P and Q are two statements.

$\implies$ – Implies.

$\sum_{n=1}^k f(n)$ – The sum from indexing $n = 1$ to $n = k$ of some function $f(n)$.

$\mathbb{Z}^+$ – The positive integers. $\mathbb{Z}$ itself represents the set of all integers.

$\binom{n}{k}$ – n choose k , a combinatorial expression that tells us how many ways we can choose

k objects from n objects assuming order doesn't matter. $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

$\square$ – *Quod erat demonstrandum*, a Latin phrase meaning “what was to be demonstrated”, marking the end of a proof.

$\operatorname{sgn} x$ – The sign of x . If x is positive, this yields 1. If it is negative, it yields -1 . Otherwise, it yields 0.

$S = \{P | Q\}$ – A set S that has elements of form P that satisfy condition(s) Q . For instance, $\mathbb{R}^+ = \{x \in \mathbb{R} | x > 0\}$.

$\int_I f \, dx$ – Integral over some interval I of $f(x)dx$. If $I = (-\infty, \infty)$, then $I = \mathbb{R}$.

$\bar{S}$ – The average value of S .

$\therefore$ – Therefore.

$\sum_i$ – Summing over i . Here, the index isn't necessarily well defined, but is case dependent and left to the reader to make discretions about the bounds of summation.

$\|\mathbf{v}\|$ – The magnitude of a vector $\mathbf{v}$. For an n dimensional vector, $\|\mathbf{v}\| = \sqrt{\sum_i^n v_i^2}$.

$\prod_{j \neq i} \mathfrak{I}_j(x)$ – The product over index j such that j doesn't equal i . This is nested within a summation over index i .

$\frac{\partial}{\partial x}$ – Partial derivative with respect to x . This is a multivariable concept, so you don't really have to worry about it.

$\psi(t, \mathbf{x})$ – A function with respect to separated space and time. This deals more with physics, so again, don't worry about it.

$\{a_n\}$ – A sequence that contains terms a_1, a_2 , and so on.

$\mathbb{N}$ – The set of natural numbers, always including $1, 2, 3, \dots$, sometimes including 0 .

$n!$ – n factorial, which is just the product of n and all positive integers less than n .

$\sup f(x)$ – The supremum of $f(x)$, its largest element.

$\mathfrak{M} : A \rightarrow B$ – A function $\mathfrak{M}$ that maps elements of set A to elements of set B . You can think of A as the domain and B as the codomain.

$\mapsto$ – Maps to.

Appendix B

This appendix contains proofs and derivations I didn't show in the rest of the text to prevent dragging out too long. It also contains some extra features of calculus that don't always appear in the first year curriculum but are nonetheless worth exploring. Those mathematically curious are recommended to peruse this chapter.

A neat result for infinite geometric series.

From algebra II, you may recall using the formula for an infinite geometric series when analyzing what values such series converge to:

$$S_G = \sum_n^{\infty} a r^n = \frac{a}{1-r},$$

where a is the initial term, and r is the common ratio. It may seem weird that an infinite series converges to something finite, but the math tells us this makes sense. Observe that

$$\begin{aligned} S_G &= a(1 + r + r^2 + r^3 + \dots) \text{ and} \\ r S_G &= a(r + r^2 + r^3 + r^4 \dots). \end{aligned}$$

Subtracting the two, we get

$$S_G - r S_G = a \left(\sum_{n=0}^{\infty} r^n - \sum_{n=1}^{\infty} r^n \right) = a.$$

We see that simply factoring the left side and rearranging the formula gives us

$$S_G = \frac{a}{1-r}$$

and we're done.

Why does $\sin^2 x + \cos^2 x = 1$? Featuring a complex explanation (literally).

We have all extensively used the trig identity

$$\sin^2 x + \cos^2 x = 1.$$

But why is it true?

We can factorize the left hand side such that

$$(\cos x + i \sin x)(\cos x - i \sin x) = 1,$$

where $i^2 \equiv -1$.

Lemma 1: $e^{ix} = \cos x + i \sin x$, $e^{-ix} = \cos x - i \sin x$.

Using this, we see that

$$(\cos x + i \sin x)(\cos x - i \sin x) = e^{ix} e^{-ix} = e^0 = 1.$$

Proving Lemma 1.

Let's write out the Taylor expansion of e^{ix} . First, we see that

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Then, we can substitute ix for x to get

$$\begin{aligned} e^{ix} &= \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\ &= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \dots \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} + i \left(\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \right) \\ &= \cos x + i \sin x. \end{aligned}$$

You can do something similar to arrive at $e^{-ix} = \cos x - i \sin x$.

The product rule for more than two functions.

In Chapter 2, we looked at the product rule, which told us

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x).$$

But what happens if we have more than two functions, namely,

$$\frac{d}{dx}(f_1(x)f_2(x)f_3(x)\dots) = \frac{d}{dx}\left(\prod_i^n f_i(x)\right)?$$

We could try brute forcing this method, but I'd rather use proof by induction. For those unfamiliar with this method of proof, we can show that a theorem is true by first showing the base case is true, then assuming the theorem is true for the n th case to prove the $(n+1)$ th case. Here, we'd like to show that the product rule for more than two functions follows

$$\frac{d}{dx}\left(\prod_i^n f_i(x)\right) = \sum_j^n \left(f'_j(x) \left[\prod_{i \neq j}^n f_i(x)\right]\right).$$

Using the base case of $n = 2$, we see that

$$\frac{d}{dx}(f_1(x)f_2(x)) = (f'_1(x)f_2(x) + f'_2(x)f_1(x)),$$

which is exactly what the product rule tells us. Now, taking the theorem to be true, we want to show

$$\frac{d}{dx}\left(\prod_i^{n+1} f_i(x)\right) = \sum_j^{n+1} \left(f'_j(x) \left[\prod_{i \neq j}^{n+1} f_i(x)\right]\right).$$

Observe that we can write

$$\prod_i^{n+1} f_i(x) = g(x)f_{n+1}(x), \text{ where } g(x) = \prod_i^n f_i(x).$$

Thus we can apply the product rule and get that

$$\frac{d}{dx}\left(\prod_i^{n+1} f_i(x)\right) = \frac{d}{dx}\left(\prod_i^n f_i(x)\right)f_{n+1}(x) + \left(\prod_i^n f_i(x)\right)f'_{n+1}(x).$$

We now rewrite the right hand side.

$$\begin{aligned} \frac{d}{dx} \left(\prod_i^n f_i(x) \right) f_{n+1}(x) + \left(\prod_i^n f_i(x) \right) f'_{n+1}(x) &= \sum_j^n \left(f'_j(x) \left[\prod_{i \neq j}^n f_i(x) \right] \right) f_{n+1}(x) + \left(\prod_i^n f_i(x) \right) f'_{n+1}(x) \\ &= \sum_j^n \left(f'_j(x) \left[\prod_{i \neq j}^{n+1} f_i(x) \right] \right) + \left(\prod_i^n f_i(x) \right) f'_{n+1}(x) \\ &= \sum_j^{n+1} \left(f'_j(x) \left[\prod_{i \neq j}^{n+1} f_i(x) \right] \right) \quad \square \end{aligned}$$

Thus, by induction, we have shown that

$$\frac{d}{dx} \left(\prod_i^n f_i(x) \right) = \sum_j^n \left(f'_j(x) \left[\prod_{i \neq j}^n f_i(x) \right] \right).$$

A look at Lagrange's error bound.

In Chapter 7, we briefly touched on the Lagrangian Error Bound. It told us that the difference between a function and its n th degree Taylor polynomial differed by no more than the maximum $(n+1)$ th derivative of $f(x)$ over the interval $[a, x]$. Mathematically, the error $R_n(x) = |f(x) - T_n(x)|$ has an upper limit

$$R_n(x) \leq \sup \left(\frac{|f^{n+1}(x)| |x - a|^{n+1}}{(n+1)!} \right).$$

We'd like to see how this is so. If we take some derivatives, we find that

$$\frac{d^{n+1}}{dx^{n+1}} R_n(x) = \frac{d^{n+1}}{dx^{n+1}} |f(x) - T_n(x)| \leq \sup (|f^{n+1}(x)|).$$

We can integrate this expression over interval $[a, x]$ to get

$$\int_a^x \frac{d^{n+1}}{dx^{n+1}} R_n(x) dx = \frac{d^n}{dx^n} R_n(x) \Big|_a^x \leq \sup (|f^{n+1}(x)|) (x - a).$$

We integrate this a few more times to observe a pattern.

$$\begin{aligned} \int_a^x \sup (|f^{n+1}(x)|) (x-a) \, dx &\geq \frac{d^{n-1}}{dx^{n-1}} R_n(x) \Big|_a^x \\ &= \sup (|f^{n+1}(x)|) \frac{(x-a)^2}{2} \\ \int_a^x \sup (|f^{n+1}(x)|) \frac{(x-a)^2}{2} \, dx &= \sup (|f^{n+1}(x)|) \frac{(x-a)^3}{2 \cdot 3}. \end{aligned}$$

We notice a pattern forming, and taking the integral $n+1$ times helps us confirm.

$$\begin{aligned} \int_a^x \sup (|f^{n+1}(x)|) \frac{(x-a)^3}{2 \cdot 3} \, dx &= \sup (|f^{n+1}(x)|) \frac{(x-a)^4}{2 \cdot 3 \cdot 4} \\ &\vdots \\ \frac{d^2}{dx^2} R_n(x) \Big|_a^x &\leq \int_a^x \sup (|f^{n+1}(x)|) \frac{(x-a)^{n-2}}{(n-2)!} \, dx = \sup (|f^{n+1}(x)|) \frac{(x-a)^{n-1}}{(n-1)!} \\ \frac{d}{dx} R_n(x) \Big|_a^x &\leq \int_a^x \sup (|f^{n+1}(x)|) \frac{(x-a)^{n-1}}{(n-1)!} \, dx = \sup (|f^{n+1}(x)|) \frac{(x-a)^n}{n!} \\ R_n(x) \Big|_a^x &\leq \int_a^x \sup (|f^{n+1}(x)|) \frac{(x-a)^n}{n!} \, dx = \sup (|f^{n+1}(x)|) \frac{(x-a)^{n+1}}{(n+1)!}. \end{aligned}$$

Making everything positive so that our error does not become negative (we only care absolute the “distance” our approximation is to $f(x)$, not whether it’s over or undershot), and recognizing that $R_n(a) = 0$, we arrive at our Lagrange error bound:

$$R_n(x) \leq \sup \left(\frac{|f^{n+1}(x)| |x-a|^{n+1}}{(n+1)!} \right).$$